QUALIFICATION STANDARD AND PLAN

Communications Information Systems and Networks Admin (CISN ADMIN)

NAV COMM 00299

NQUAL – AISN

MITE ID
Mod 1 – 123169
Mod 2 – 123170

RECORD OF AMENDMENTS

1. The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

2. Qualification Standard and Plan (QSP) versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents. Versions with a number followed by a bracketed number indicate a stray from this nomenclature. The number in brackets indicates the corrected version number.

Version Number	Amendment Type	Approval Date	Authority
3.0	QSP Creation	14/12/2021	NTDC(A)
3.0	Quality control	8/8/2023	CNTG

TABLE OF CONTENTS

RECORD OF AMENDMENTS ..... **I**

TABLE OF CONTENTS ..... **II**

FOREWORD ..... **IX**

CHAPTER 1: TRAINING MANAGEMENT DETAILS **1**

AIM 2
COURSE DURATION 2
REQUIRED COURSE STAFFING 2
COURSE CAPACITY 3
PREREQUISITES 3
PRIOR LEARNING ASSESSMENT AND RECOGNITION 3
SPECIAL INSTRUCTIONS 3

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE **4**

PO 001 – DRAFT COMMUNICATIONS MESSAGES 5
EO 001.01 – Plan and Conduct Communications Exercise and Training 6
LG 001.01.01 – Drafting Communication Exercise Messages 6
EC 001.01 – Plan and Conduct Communications Exercises and Training 8
EO 001.02 – Report Communication Status 11
LG 001.02.01 – Report Network Disruptions and Outages 11
LG 001.02.02 – Draft Communication Status Messages 13
LG 001.02.03 – Provide Input to Deployment Reports 14
LG 001.02.04 – Draft Communication Operational Messages 15
EC 001.02 – Report Communication Status 16
PC 001 – Draft Communications Messages 20
PO 002 – DIRECT CCR OPERATIONS 21
EO 002.01 – Direct Flash-up and Shutdown of Communications Equipment 23
LG 002.01.01 – Identify RADHAZ Policy and Procedure 23
EC 002.01.01 – Identify RADHAZ Policy and Procedures 26
LG 002.01.02 – Manage Pre-Sail Checks 27
EC 002.01.02 – Manage Pre-Sail Checks 29
EO 002.02 – Direct Transmission and Reception Information by Communications Circuits 30
LG 002.02.01 – Direct Transmission and Reception of Information by Communication Circuits 30
EC 002.02 – Direct Transmission and Reception Information by Communications Circuits 32
EO 002.03 – Direct SAR Communication Procedures 33
LG 002.03.01 – Manage CCR and Bridge SAR Checklist 33
EC 002.03 – Direct SAR Communication Procedures 35
PC 002 – Direct CCR Operations 36
PO 003 – DIRECT VISUAL COMMUNICATIONS 37
EO 003.01 – Conduct Duties of Flag Deck Supervisor 38
LG 003.01.01 – Entering and Leaving Harbour Anchoring/Mooring Flag India Procedures 38
EC 003.01.01 – Entering and Leaving Harbour Anchoring/Mooring Flag India Procedures 40
LG 003.01.02 – RAS flag Deck Procedures 41
EC 003.01.02 – RAS Flag Deck Procedures 43
PC 003 – Direct Visual Communications 44
PO 004 – ADMINISTER SERVER VIRTUALIZATION 45
EO 004.01 – Build Ship Servers to an Operational State 47
LG 004.01.01 – Installing ESXi 6.5 47
LG 004.01.02 – Create a Virtual Machine Template 48

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 004.01 – Build Ship Servers to an Operational State 48
EO 004.02 – Implement vCenter Software 52
LG 004.02.01 – Install vCenter 52
EO 004.03 – Utilizing Datastores 52
LG 004.03.01 – Utilize datastores 52
EO 004.04 – Configure vSwitches and VMware Roles and Permissions 54
LG 004.04.01 – Configure vSwitches and VMware roles and permissions 54
EC 004.04 – Configure vSwitches and VMware roles and permissions 54
EO 004.05 – Create Domain Controllers and Join the Domain 58
LG 004.05.01 – Create a domain controller 58
EC 004.05 – Create domain controllers and join the domain 59
PC 004 – Administer Server Virtualization 62
PO 005 – ADMINISTER SERVER MANAGEMENT 65
EO 005.01 – Administering DNS Server Role 66
LG 005.01.01 – Adding the DNS Server Role 66
EC 005.01 – Adding the DNS Server Role 67
EO 005.02 – Managing the Printer Service 69
LG 005.02.01 – Installing the Print Service 69
EC 005.02 – Managing the Printer Service 70
EO 005.03 – Creating Scripts and Profiles 72
LG 005.03.01 – Creating Scripts and Profiles 72
EC 005.03 – Creating Scripts and Profiles 73
PC 005 – Administer Server Management 75
PO 006 – ADMINISTER MICROSOFT EXCHANGE SERVER 77
EO 006.01 – Maintaining The Exchange Server 79
LG 006.01.01 – Configuring ECP 79
LG 006.01.02 – Configuring Mailboxes 80
LG 006.01.03 – Configuring Client Access and OWA 81
LG 006.01.04 – Configuring Transport Rules 82
LG 006.01.05 – Configuring Groups, Address List and RBAC 83
LG 006.01.06 – Installing the Exchange Management Tool on a Workstation 84
EC 006.01 – Maintaining The Exchange Server 85
PC 006 – Administer Microsoft Exchange 87
PO 007 – ADMINISTER INFORMATION SYSTEMS (IS) 89
EO 007.01 – Administer IS Troubleshooting Procedures 90
LG 007.01.01 – Conduct Fault Finding 90
LG 007.01.02 – Regulate IS Trouble Ticket 92
EC 007.01 – Administer IS Troubleshooting Procedures 94
EO 007.02 – Administer IS Equipment 97
LG 007.02.01 – Manage Workstation and Server Infrastructure 97
LG 007.02.02 – Conduct IS Status Brief 99
EC 007.02 – Administer IS Equipment 100
PC 007 – Administer Information Systems 102

CHAPTER 3: ASSESSMENT OF TRAINEE 105

GENERAL 106
ENABLING AND PERFORMANCE CHECKS 106
RESPONSIBILITY FOR TESTING 106
PROGRESS MONITORING 106
TRAINING REVIEW BOARD 107
COURSE REPORTS 107

ANNEX A – REFERENCES 1

ANNEX B – TASK LIST **1**
ANNEX C – RECORD OF DECISIONS **1**
ANNEX D – TASK ANALYSIS DIAGRAM **24**
ANNEX E – COSTING INFORMATION **1**
ANNEX F – ASSESSMENT PLAN **1**
ANNEX G – TEL DEVELOPMENT OPPORTUNITIES **1**
ANNEX H – SCHEDULING INFORMATION **1**

Acronyms and Abbreviations

A	
AAR	After Action Report
ART	Article
ATP	Allied Tactical Publication
ATT	Actual Training Time
AXP	Allied Exercise Publication
B	
BAT	Base Administrative Time
C	
CAF	Canadian Armed Forces
CCR	Communications Control Room
CFCD	Canadian Forces Controlled Document
CFITES	Canadian Forces Individual Training and Education System
CIS	Communication and Information Systems
Civ	Civilian
Cmdt	Commandant
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
COMSPOT	Communication Spot
COMSTAT	Communication Status
CSNI	Consolidated Secret Network Infrastructure
D	
DAOD	Defence Administrative Orders and Directives
DC	Domain Controller
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DNS	Domain Name System
DPGR	Director Personnel Generation Requirements
DWAN	Defence Wide Area Network
E	
EC	Enabling Check
EDI	Explain Demonstrate Imitate
EO	Enabling Objective
EPIRB	Emergency Position Indicating Radio Beacon
ETR	Estimated Time of Repair
EXSi	Elastic Sky X Integrated
F	
FBB	Fleet Broad Band
G	
GPNet	General Purpose Network
H	
HERO	Hazards of Electromagnetic Radiation to Ordnance
HERP	Hazards of Electromagnetic Radiation to Personnel
HF	High Frequency

HMCS	His Majesty's Canadian Ship
HOD	Head of Department
I	
IAW	In Accordance With
IC	In Charge
ID	Identification
IMS	Information Management Standard
IMSRT	Intermediate Multi Ship Readiness Training
IS	Information Systems
ISA	Information Systems Administrator
IT	Information Technology
L	
LO	Learning Objective
LG	Lesson Guidance
LUN	Logical Unit Number
M	
MI	Master Index
MITE	Military Individual Training and Education
MOB	Man Overboard
MOSID	Military Occupational Structure Identification Code
MS	Master Sailor
MSO	Message Signal Operator
N	
NCITT	Naval Communicator Information Technology Trainer
NCS	Naval Communication Station
NIL	Nihil
NFS	Naval Fleet School
NRS	Naval Radio Station
NavIS	Navy Information System
NAVORD	Naval Order
NMT	Navy Multiband Terminal
NOC	Network Operation Center
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order
NQual	Naval Qualification
NTDC(A)	Naval Training Development Centre (Atlantic)
NTFCR	Non Tactical Frequency Clearance Request
O	
OCS	Officer Conducting the Serial
OJT	On-the-Job Training
OJPR	On-the-Job Performance Requirement
OPTASK	Operational Tasking
ORS	Operations Room Supervisor
OVF	Open Virtualization Format

OWA	Outlook Web Access
P	
PC	Performance Check
PIM	Promulgated Intended Movement
PLAR	Prior Learning Assessment and Recognition
PO	Performance Objective
PO1	Petty Officer First Class
PO2	Petty Officer Second Class
PRC	Portable Radio Communications
PT	Physical Training
Q	
QL	Qualification Level
QSP	Qualification Standard and Plan
R	
RADHAZ	Radiation Hazard
RAS	Replenishment at Sea
RATT	Radio Teletypewriter
RF	Radio Frequency
R&Q	Rations & Quarters
RCN	Royal Canadian Navy
S	
S1	Sailor First Class
SAA	Satellite Access Assignment
SAR	Search and Rescue
SART	Search and Rescue Transponder
SAT	Satisfactory
SATCOM	Satellite Communication
SHOW	Senior Hand of the Watch
SO	Standing Order
SOC	Satellite Operations Centre
SOP	Standard Operating Procedure
S/S	Ship/Shore
SSOs	Ship's Standing Orders
T	
TABORD	Table Order
TAT	Training Administrative Time
TA	Training Authority
TD	Temporary Duty
TE	Training Establishment
TEL	Technology Enabled Learning
TG Tac	Task Group Tactical
TGOA	Task Group Operation Administration
TO	Training Officer
TPI	Two Person Integrity
TPO	Training Petty Officer

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

TRB	Training Review Board
TS	Training Site
TTT	Training Travel Time
U	
UNSAT	Unsatisfactory
V	
VHF	Very High Frequency
VM	Virtual Machine
Vol	Volume
W	
W&SB	Watch & Station Bill
WGS	Wideband Global Satellite
WSOC	Wideband SATCOM Operation Center

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

FOREWORD

1. This training is based on the specification found at Director Personnel Generation Requirements (DPGR) website: **http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page**.

   a. Tasks are based on the master task list that was submitted to DPGR in February 2020.

2. A trainee will be granted the National Qualification (NQual) AISN when the standards listed in all POs have been met.

3. A trainee will be granted the Military Individual Training and Education (MITE) identifications (IDs) when standards listed in all POs have been met:

   a. Module 1: PO 001 through to PO 003 - 123169

   b. Module 2: PO 004 through to PO 007 - 123170

4. The designated TEs for this training include all NFSs within NPTG.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

AIM

1. The aim of this QSP is to establish the requirements and training for the CISN Administrator (AISN). This QSP is designed to assist NTDC(A) in developing, NFSs in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) Naval Personnel and Training Order (NPTORD) 4500-22 Annex B Table 3.1. Partial days are rounded up. For this course, a period of instruction is 50 minutes in duration and each day has 8 instructional periods. The course duration calculation is based on maximum load.

Course Duration Calculation		
Allocation of Actual Training Time (ATT)	Periods	Days
Lesson Guidance (LG)	180	23
Enabling Checks (ECs)	37	5
Performance Checks (PCs)	20	3
ATT SUB TOTAL:	**237**	**31**
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	9	2
Training Administrative Time (TAT)	9	2
Training Travel Time (TTT)		
Physical Fitness (PT)	20	3
BAT, TAT, TTT, PT SUB TOTAL:	**38**	**7**
(ATT+BAT+TTT+TAT+PT) **TOTAL:** 400 minutes	**275**	**38**

REQUIRED COURSE STAFFING

3. The following personnel, including guest lecturers, are required to conduct this training:

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Course Director	PO2	1	QL5B	SECRET	All
Senior Instructor	PO1	1	QL6B	SECRET	All
Instructor	MS	1	PLQ/QL5B	SECRET	All
Technologist	Civ	1	Technologist	Reliability	PO 4 - 7

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

COURSE CAPACITY

4. Planning for the conduct of training is based on availability of resources; course capacities are as follows:

5. Maximum – 12;

6. Minimum – 6; and

7. Optimum – 8.

PREREQUISITES

8. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

9. NAV COMM CISN Operator (AIXC) National Qualification;

10. A level II security clearance prior to course commencement; and

11. Trainee consolidation should be initiated by the TDC and confirmed by the individual's Unit Training Coordinator prior to course loading.

PRIOR LEARNING ASSESSMENT AND RECOGNITION

12. Prior Learning Assessment and Recognition (PLARs) shall be conducted IAW Naval Order (NAVORD) 4500-2, Defence Administrative Orders and Directives (DAOD) 5031-1, and Canadian Forces Individual Training and Education System (CFITES) Volume (Vol) 12.

SPECIAL INSTRUCTIONS

13. This course is modularized to facilitate flexibility in the delivery of training to NAV COMMs. In order for NAV COMM CISN ADMIN to receive the competency code AISN, they will require both MITE codes.

14. A trainee will be granted the Military Individual Training and Education (MITE) identifications (IDs) when standards listed in all POs have been met:

   a. Module 1: PO 001 through to PO 003 - 123169

   b. Module 2: PO 004 through to PO 007 - 123170

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 001 – DRAFT COMMUNICATIONS MESSAGES

1. **Performance**: Draft Communications Messages.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) CSNI Workstations, and
      (3) Printer.

   b. **Denied**: nil; and

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall draft communications messages by:

   a. Planning and conducting communications exercises and training by:

      (1) Drafting Communication Exercise Messages (GT0200).

   b. Reporting communication status to by:

      (1) Reporting Network Disruptions and Outages (ET0470);
      (2) Drafting Communication Status Messages (DT0390);
      (3) Providing Input to Deployment Reports (DT0480); and
      (4) Drafting communication operational messages.

4. **Training Limitations**: Nil.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 001.01 – PLAN COMMUNICATIONS EXERCISE AND TRAINING
LG 001.01.01 – DRAFTING COMMUNICATION EXERCISE MESSAGES

LG 001.01.01 – Drafting Communication Exercise Messages		
Location Conditions		**Instructional Sequence**
Classified Classroom		Beginning of course
Time	**Classification**	**Instructor**
400 minutes	Confidential	MS Nav Comm or above
Learning Objectives		**Instructional Method(s)**
1) Training Messages a. TABORDs b. OCS INTENTIONS c. AAR		1) Guided Discussion 2) EDI 3) Individual Study
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to plan and conduct communications exercises and training.	
Activity Steps	1) Instructor is to give each trainee a sample message handout and AXP 3. 2) Instructor is to open the lesson with a Guided Discussion, asking trainees what their past experiences are with drafting TABORD, AARs, and OCS INTENTIONS messages. 3) Instructor is to review AXP 3 table 1-1 pg. 1-4/5 with the class (Line number assignments for TABORDS) using the TABORD message example as a guide. Instructor is to go through the table line per line explaining the contents of said lines amplified with TABORD message example. 4) Instructor will demonstrate drafting a TABORD message on a projector from a scenario provided, utilizing a TABORD message template and exercises derived from AXP 3 as well as fictitious unit/timing information. 5) Trainees will then perform drafting a TABORD message from a scenario provided utilizing a TABORD message template and exercises derived from AXP 3 as well as fictitious unit/timing information of their own choosing. 6) Instructor will then use the TABORD drafted with the class to draft an Exercise Report (AAR) using AXP 3 D as a reference and guide, reviewing line contents with the trainees. 7) Trainees will then perform drafting an AAR utilizing their own previously drafted TABORD exercise. 8) Instructor is to have trainees practice drafting TABORD and Exercise Report (AAR) messages from practice scenarios provided. Once complete, messages are to be reviewed as a class.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 001.01.01 – Drafting Communication Exercise Messages	
Instructor Resources	**References**
1) Message Template Handouts 2) OPTASK COMMS 3) TABORD Exercise Scenarios	1) AXP 3
Note(s) to Instructor	
Nil.	

EO 001.01 – PLAN AND CONDUCT COMMUNICATIONS EXERCISES AND TRAINING

EC 001.01 – PLAN COMMUNICATIONS EXERCISES AND TRAINING

EC 001.01 – Plan Communications Exercises and Training		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.01	Classified Classroom	After LG 001.01.01
Time	**Classification**	**Instructor**
100 minutes	Confidential	MS or above Nav Comm
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to plan and conduct communications exercises and training.	
Assessment Steps	**Input – Part 1:** Trainees will be given a scenario of a unit/ship at sea conducting an exercise. Trainee will be expected to draft a TABORD exercise message from the scenario provided using an exercise designated by the instructor. **Part 2:** Trainees will then be required to draft an Exercise Report (AAR) using their previously drafted TABORD. **Performance –** Trainees will have 100 minutes to complete this assessment. **Output –** Trainee must attain 13/13 to pass this assessment. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) Sample Message Handout 2) EC Scenarios 3) OPTASK COMMS		
Note(s) to Instructor		
1) Instructor is to designate exercises used within AXP 3 for this EC. Exercise Report (AAR) must be drafted based on the trainees drafted TABORD. 2) Trainees are allowed the use of refs and resources during this EC.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 001.01 Plan Communications Exercises and Training			
Item	**SAT**	**UNSAT**	**Remarks**
1. Draft a TABORD			
A. Classification	☐	☐	
B. Addressees:			
a. TO (ACTION ADDRESSEES)	☐	☐	
b. FM (STJ)	☐	☐	
C. OCS - STJ	☐	☐	
D. D. START TIME	☐	☐	
E. E. LENGTH OF EXERCISE (max 2 HRS)	☐	☐	
F. F. EXERCISE INSTRUCTIONS AS PER AXP 3	☐	☐	
G. G. ACKNOWLEDGEMENT REQUIRED	☐	☐	
2. Draft AAR based on TABORD			
A. CLASSIFICATION	☐	☐	
B. ADDRESSEES			
TO	☐	☐	
FM	☐	☐	
C. SUBJECT	☐	☐	
D. ACKNOWLEDGEMENT REQUIRED	☐	☐	

Score: /13

Trainee must achieve 13/13 SATs to pass this assessment.

Positive Feedback	**Improvement Feedback**

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 001.02 – REPORT COMMUNICATION STATUS
LG 001.02.01 – REPORTING NETWORK DISRUPTIONS AND OUTAGES

LG 001.02.01 – Reporting Network Disruptions and Outages		
Location Conditions		**Instructional Sequence**
Classified Classroom		After EC 001.01
Time	**Classification**	**Instructor**
400 minutes	Confidential	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) CIS COMSPOT		1) Guided Discussion 2) Lecture 3) Demonstration 4) Performance
Scenario	As a SHOW onboard HMCS HALIFAX, you will be responsible for responding to and reporting network disruptions/outages.	
Activity Steps	1) Instructor to open the lesson with a discussion of the trainee's experiences with drafting/troubleshooting network outages. 2) Trainees are given CFCD 107 to read Ch.5 Section 4 pg. 5 – 11/13 Art 42- 46. 3) Instructor is to lead a lecture reviewing CIS COMSPOTS with CIS COMSPOT/REPLY/SITREP/RECT example message IAW CFCD 107 instructions. Instructor is to emphasis timings related to CIS COMPSOT including: • Timing of Initial report • Timing till drafting of Situation Report (SitRep) 4) Instructor is to lead guided discussion on the steps taken when faced with a Network/Service Outage to include: • Identify specific equipment failures or lack there of • Attempt to troubleshoot (Liaise with Tech Support) • Identify specific failed component (Piece that requires repair)	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 001.02.01 – Reporting Network Disruptions and Outages	
	• ETR • Transcribe into message (timings for messages to go out) • Rectification 5) Trainees are to draft a CIS COMSPOT from a scenario provided to the class by the instructor. Trainees shall use CFCD 107 as well as example CIS COMSPOT message as references. 6) Instructor is to end the lesson by demonstrating the drafting of the correct CIS COMSPOT from the scenario provided to the class, correcting any errors made as well as answering any questions raised by the trainees. 7) Instructor is to have trainees practice drafting CIS COMSPOT messages from practice scenarios provided. Once complete, messages are to be reviewed as a class.
Instructor Resources	**References**
1) Message Template Handouts 2) CIS COMSPOT Scenarios	1) CFCD 107
Note(s) to Instructor	
Nil.	

EO 001.02 – REPORT COMMUNICATION STATUS
LG 001.02.02 – DRAFTING COMMUNICATION STATUS MESSAGES

LG 001.02.02 – Drafting Communication Status Messages		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.02.01	
Time	**Classification**	**Instructor**
400 minutes	Confidential	MS or above Nav Comm
Learning Objectives	**Method(s)**	
1) COMSTAT 2) COMMSWHEREABOUTS 3) WGS Report	1) Lecture 2) Guided Discussion 3) Floating Flipchart	
Scenario	As a SHOW onboard HMCS HALIFAX, you will be required to draft communication status messages.	
Activity Steps	1) Instructor to initiate the lesson with a Floating Flipchart exercise. Instructor is to place 3 flipchart pages on the walls (1 for COMSTAT, COMMSWHEREABOUTS and WGS Report). Instructor will have the trainees write examples of when these messages would be used or how they've been used within their experiences. 2) Instructor is then to review the flipcharts in with the class, expanding on the uses of each message and their role to Nav Comms. 3) Trainees are then given CFCD 107 to read Ch. 2 Section 2 pg. 2 – 4. 4) Instructor is to review material read with the class. 5) Instructor is to lead a guided discussion on SATCOM Daily SitRep. 6) Trainees are to review CFCD 107 Ch. 7 Section 7 pg. 7 – 5/6 Art 19 – 21. (SATCOM Daily SitRep) 7) Instructor is to lead a guided discussion reviewing the material covered with the trainees, letting the trainees share their experiences with SATCOM Daily SitReps or PIMS data woes. Instructor will lead a lecture breaking down the information and emphasizing key information, including: • Frequency of the SitReps • What is PIMS data/Where to find PIMS data • Importance of accurate PIMS data	
Instructor Resources	**References**	
1) Message Template Handouts 2) Floating Flipcharts	1) CFCD 107	
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 001.02 – REPORT COMMUNICATION STATUS
LG 001.02.03 – PROVIDING INPUT TO DEPLOYMENT REPORTS

LG 001.02.03 – Providing Input to Deployment Reports		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.02.02
Time	**Classification**	**Instructor**
200 minutes	Confidential	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) SATCOM AAR		1) Guided Discussion 2) Demonstration
Scenario	As a SHOW onboard HMCS HALIFAX, you will be required to provide input to deployment reports.	
Activity Steps	1) Trainees are given CFCD 107 to read Ch. 7, section 7, pg. 7 – 6/7, Art 22. 2) Instructor to lead a guided discussion reviewing the material read from CFCD 107. Instructor is to lead an open discussion on tactics sections can use to record information for deployment reports, to include: • Lessons Learned binder • CIS COMSPOT repository • NMT fault logs 3) Instructor is to review SATCOM AAR instructions using SATCOM SAA message as a template and demonstrate drafting an example SATCOM AAR as a class. 4) Instructor is to review key authorities and their roles pertaining to the SATCOM AAR and its importance to future missions to include: • 76 Comms group • SOC and WSOC • NRS • NOC	
Instructor Resources		**References**
1) SATCOM SAA 2) Message Template Handouts		1) CFCD 107
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 001.02 – REPORT COMMUNICATION STATUS
LG 001.02.04 – DRAFTING COMMUNICATION OPERATIONAL MESSAGES

LG 001.02.04 – Drafting Communication Operational Messages		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.02.03
Time	**Classification**	**Instructor**
400 minutes	Confidential	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Tactical and Non-Tactical Frequency Clearance Requests 2) ZIC – ZID Message		1) Lecture 2) Guided Discussion 3) Demonstrate 4) Performance
Scenario	As a SHOW onboard HMCS HALIFAX, you will be required to draft communication operational messages.	
Activity Steps	1) Instructor is to introduce the lesson with a Guided Discussion on trainee's familiarity with Tactical/Non-Tactical Frequency Clearance Requests. Instructor is to also briefly summarize the use of a FINAL ZID message to MSOs and their SHOWs. 2) Trainees are given CFCD 107 to review Ch. 4, Section 4 pg. 4 – 9/11, Art 43 – 51. Afterward, the instructor is to lead a review of the covered articles utilizing a Tactical Frequency Clearance Request (TFCR) template and CIB 009/12 as a guide. Instructor is to also review timings associated with Non-Tactical Frequency Clearance Request (NTFCR) as well as authorities involved. 3) Trainees are then to draft a Tactical Frequency Clearance Request message from a scenario provided. Trainees may use TFCR template and CIB 009/12 as a guide. 4) Instructor is then to demonstrate drafting the TFCR message from the scenario provided with the class, correcting any errors trainees may have made as well as answering any questions raised by the trainees. 5) Instructor is to have trainees practice drafting NTFCR messages from practice scenarios provided. Once complete, messages are to be reviewed as a class.	
Instructor Resources		**References**
1) Message Template Handouts 2) NTFCR Exercise Scenarios		1) CFCD 107 2) CIB 009/12
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 001.02 – REPORT COMMUNICATION STATUS
EC 001.02 – REPORT COMMUNICATION STATUS

EC 001.02 – Report Communication Status		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.02	Classified Classroom	After LG 001.02.04
Time	**Classification**	**Instructor**
200 minutes	Confidential	MS or above Nav Comm
Scenario	As a SHOW onboard HMCS HALIFAX, you will be required to report communication status.	
Assessment Steps	**Input – Part 1: (For scenarios, 1 different platform for each) Trainees** will be given a scenario of a ship expected to depart for Norfolk, USA and another scenario of a ship departing for Rota, Spain. Trainees will be required to draft a Tactical Frequency Clearance Request message for each scenario provided. **Part 2:** Trainees will be given a scenario of a ship/unit out at sea experiencing a network outage. Trainees will be required to draft a CIS COMSPOT message from the scenario provided. **Part 3:** Trainees will begin the exercise with a scenario providing them with details required to draft a WGS Daily SitRep. **Performance –** Trainees will have 200 minutes to complete this assessment. **Output –** Trainee must attain 27/27 to pass this assessment. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) Sample Message Handout 2) EC Scenarios		
Note(s) to Instructor		
1) Trainees are allowed the use of refs and resources during this EC.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 001.01 Report Communication Status			
Item	**SAT**	**UNSAT**	**Remarks**
1. Draft a Non-Tactical Frequency Clearance Request			
A. Classification	☐	☐	
B. Date	☐	☐	
C. Addressees			
TO	☐	☐	
FM	☐	☐	
D. Port	☐	☐	
E. TFCR Dates required	☐	☐	
F. Emissions Required			
HF	☐	☐	
VHF	☐	☐	
UHF	☐	☐	
Ka Band	☐	☐	
2. Draft a Network Outage Report			
A. Classification - C	☐	☐	
B. Addressees			
TO	☐	☐	
FM	☐	☐	
C. Priority – PRI 1	☐	☐	
D. Services affected - DWAN	☐	☐	
E. Troubleshooting steps (SOC, WX etc)	☐	☐	
F. Time	☐	☐	
3. Draft a SATCOM Daily SitRep – WGS Report			
A. Satellite Area – 230600ZMAY21	☐	☐	
B. TERM TX Freq – HAL 8123.45	☐	☐	
C. TPO – 5W	☐	☐	
D. Distant End RVC LVL – 3.9 GMT LMAA-HFX -50	☐	☐	
E. BER – 2.2 x 10-7	☐	☐	
F. TRK – Auto-track	☐	☐	
G. WX - CLR	☐	☐	
H. PIMS (all five included)	☐	☐	
I. Terminal Operator's Initials (trainee's)	☐	☐	
J. « END OF REPORT »	☐	☐	
Score: /27			

The trainee must achieve 27/27 SATs to pass this assessment.			
Positive Feedback		**Improvement Feedback**	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 001 – DRAFT COMMUNICATIONS MESSAGES
PC 001 – DRAFT COMMUNICATIONS MESSAGES

PC 001 – Draft Communications Messages		
Scope	**Location Conditions**	**Instructional Sequence**
PO 001	Classified Classroom	After EC 001.02
Time	**Classification**	**Instructor**
100 minutes	Confidential	MS or above Nav Comm
Scenario	As a SHOW onboard HMCS Halifax, you will be required to draft communications messages.	
Assessment Steps	**Input –** Trainees are given a 15-question written assessment. **Performance –** Trainees are given 100 minutes to complete this EC **Output –** Trainees must answer 12/15 questions correctly to pass this EC **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) PC		
Note(s) to Instructor		
1) No refs and/or resources are to be used during this PC.		
Assessment Limitations	Nil.	
Impact of Failure	IAW NPTORD 4500-54.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 002 – DIRECT CCR OPERATIONS

1. **Performance**: Direct Communications Control Room (CCR) Operations.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) DWAN Workstation; and
      (3) Printer.

   b. **Denied**: Nil.

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall direct Communications Control Room operations by:

   a. Directing the Flash-Up and Shut Down of Communications Equipment (DT0030) by:

      (1) Identifying RADHAZ policy and procedures by:
         (a) Identifying HERO and HERP considerations; and
         (b) Responding to RADHAZ incidents.
      (2) Managing pre-sail checks by:
         (a) Managing equipment flash-up and shut down; and
         (b) Directing communication checks.

   b. Directing Transmission and Reception of Information by Communications Circuits (DT0190) by:

      (1) Maintaining message handling procedures;
      (2) Directing new day procedures;
      (3) Supervising communications checks;
      (4) Monitor Military Satellite Connectivity (BT0030);
      (5) Monitor Civilian Satellite Connectivity (BT0040);
      (6) Troubleshoot Military Satellite Connectivity (BT0050);
      (7) Troubleshoot Civilian Satellite Connectivity (BT0060);
      (8) Enforce Compliance with River City State (FT0380);

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

(9) Contribute to Emergency Destruction Plan (FT0160);
(10) Destroy Cryptographic Equipment (FT0170); and
(11) Destroy Communication Equipment (FT0180).

c. Directing search and rescue (SAR) communication procedures to include:

(1) Managing CCR and Bridge SAR checklist to include:
(a) Implementing River City procedures.

4. **Training Limitations**: Nil.

EO 002.01 – DIRECT FLASH-UP AND SHUTDOWN OF COMMUNICATIONS EQUIPMENT
LG 002.01.01 – IDENTIFYING RADHAZ POLICY AND PROCEDURE

LG 002.01.01 – Identifying RADHAZ Policy and Procedure		
Location Conditions		**Instructional Sequence**
Classroom		After PC 001
Time	**Classification**	**Instructor**
500 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Identify RADHAZ Policy and Procedures 2) Response to RADHAZ Incident 3) Identify HERO and HERP Considerations		1) Guided Discussion 2) Video 3) Think-Pair-Share 4) Best Summary
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to identify RADHAZ policy and procedures to ensure personnel safety.	
Activity Steps	1) Instructor to lead a guided discussion drawing from the trainees' personal experiences with regards to the following: • RADHAZ incident and response • MOB RADHAZ considerations • RAS RADHAZ considerations • Flying Stations RADHAZ considerations • HERO and HERP 2) Instructor is to show a 15 mins video on RF Awareness Safety. 3) Instructor is to have trainees review RCN RF Safety Program as described in SEMs: • RF Incident reporting procedures, SOP-S09 Art 84 - 89 • Rescue Stations, SOP-S09B3 • RAS RF procedures, SOP-S09 Art 24-25 • Flying Stations considerations, SOP-S09 Art 26, SOP-S09B1 & SOP-S09B1A • Normal Control procedures, SOP-S09 Art 36-39 4) Instructor is to lead a review of the material covered from RCN RF Safety directive. 5) Instructor is to facilitate a Think-Pair-Share activity. Using the topics listed below, trainees are to be divided into X groups, with X topics per group. Each group is to use SEMs to familiarize themselves with their topics and create a presentation to give to the class. Each to represent a different ship's platform. X representing the variable – instructor to decide numbers based on course loading. • Antennae locations • Types of emitters • HF power levels		

LG 002.01.01 – Identifying RADHAZ Policy and Procedure

- RF warning sign and location
- Areas out-of-bounds

a) HARRY DEWOLF SEMS REFS

- Appendix SOP-S09B1-1/3 – 3/3 HERP Uncontrolled Environment Restrictions
- Appendix SOP-S09B5 RF Warning Sign Placement for HDW Class Ships (permanent and temporary)

b) HALIFAX CLASS SEMS

- ANNEX SOP-S09B1 - HERP UNCONTROLLED ENVIRONMENT RESTRICTIONS FOR HALIFAX CLASS SHIPS (page SOP-S09B1-1/6 to 6/6)
- Table SOP-S09B5.3 – Temporary RF Warning Sign Locations for Post- MLR HALIFAX Class Ships at CE (pg. SOP-S09B5-5/8)

c) KINGSTON CLASS SEMS

- ANNEX SOP-S09B1, Table SOP-S09B1.1 - HERP Uncontrolled Environment Restrictions for KINGSTON Class (QETE C040015) (pages SOP-S09B1-1/2 and SOP-S09B1-2/2)
- APPENDIX SOP-S09B5 RF warning sign placement for KINGSTON class ships (page SOP-S09B5-1/1)

d) VICTORIA CLASS SEMS

- ANNEX S9B –PERSONNEL ALOFT PRECUATIONS
- Instructor to note that as submarines are not required to surface to send and receive message traffic thus personnel would not be on the fin; when surfaced personnel are not to be on the bridge until transmissions have ceased and they are RADHAZ safe.

6) Instructor to facilitate Team Quiz activity. In this activity, instructors to call out a class of ship & power level. Teams to use SEMs in order to identify the out-of-bounds areas.

7) Instructor is to have trainees conduct AIT Activity Think-Pair-Share/Best Summary. Trainees will be broken up into groups of 2 and provided with 3 scenarios. With the scenarios provided, trainees will determine the out-of-bounds areas for each power level using SEMs as a reference. Once completed, each group member will present their scenario breakdown of out-of-bounds areas for their ships' power level.

8) Once the exercise is completed and each trainee has presented what they've learned, the instructor is to provide feedback.

LG 002.01.01 – Identifying RADHAZ Policy and Procedure	
Instructor Resources	**References**
1) 20210419-EO_002.01.01_RF_Safety_Awareness 2) 20210818-EO_002.01.01_Out_of_Bounds_Areas_Scenario_Activity_Step_6 3) 20210818-EO_002.01.01_CPF Antenna diagram – Comms 4) 20210818-EO_002.01.01_CPF Antenna diagram 5) 20210818-EO_002.01.01_CSS_Antenna_Arrangement 6) 20210818-EO_002.01.01_Harry_DeWolf-class 7) 20210818-EO_002.01.01_HDW_Antenna_Arrangement 8) 20210818-EO_002.01.01_KIN Antenna diagram 9) 20210818-EO_002.01.01_KIN Antenna Diagram2 10) 20210818-EO_002.01.01_MCDV 11) 20210922-EO_002.01.01_RADHAZ Zones QETE Simulation 12) 20210823-EO_002.01.01_Submarine_Diagram	1) HAL_SEMS_June21 2) HDW_Class_SEMS_OP_DRAFT_NOV20 3) KIN_SEMS_May21 4)MARPAC_305740_v15_Victoria_Class_SEMS_Manual
Note(s) to Instructor	
Nil.	

EO 002.01 – DIRECT THE FLASH UP AND SHUT DOWN OF COMMUNICATIONS EQUIPMENT
EC 002.01.01 – IDENTIFY RADHAZ POLICY AND PROCEDURES

EC 002.01.01 – Identify RADHAZ Policy and Procedures		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.01.01	Classroom	After LG 002.01.01
Time	**Classification**	**Instructor**
50 minutes	Unclassified	MS Nav Comm or above
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct RADHAZ policies and procedures.	
Assessment Steps	**Input:** Trainees will be given a 10-question written assessment that requires them to demonstrate their knowledge with identifying RADHAZ policies and procedures. **Performance –** Trainees will have 50 minutes to complete this assessment. **Output –** Trainee must attain 8/10 SAT to pass this assessment. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20210818-EC_002.01.01		
Note(s) to Instructor		
Trainees are permitted to use SEMs as well as any personal notes they may have taken to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 002.01 – DIRECT THE FLASH UP AND SHUT DOWN OF COMMUNICATIONS EQUIPMENT
LG 002.01.02 – MANAGING PRE-SAIL CHECKS

LG 002.01.02 – Managing Pre-Sail Checks		
Location Conditions		**Instructional Sequence**
Classroom		After EC 002.01.01
Time	**Classification**	**Instructor**
400 minutes	Unclassified	MS and above Nav Comm
Learning Objectives		**Method(s)**
1) Manage Pre-sail Checks 2) Manage Equipment Flash-Up and Shutdown 3) Direct communication checks		1) Floating Flipchart 2) Guided Discussion
Scenario	As a SHOW onboard HMCS HALIFAX, you will be directing a CCR flash-up and pre-sail checks.	
Activity Steps	1) Instructor is to introduce the lesson with a Floating Flipchart exercise. One flipchart placed on one side of the room will be labelled "Pre-Sails". On the other side of the room, a flipchart will be labelled “Flash-Up/Shutdown”. Trainees will be asked to individually write what they believe are the duties of a SHOW preparing a ship to go to sea as related to the flipchart topic names. Once each trainee has contributed at least one item, the instructor is to review the results with the class. 2) Instructor is to lead a guided discussion on the various actions a SHOW would direct before leaving for a sail. The instructor is to allow for input from trainees. Topics to include but not limited to: • Positional Requirements (W&SB, personnel required for flash-up, Bridge Team, Flag Deck team, CCR Team, MSO) • Ensuring PRCs are prepped (MOB PRCs/Bridge PRCs • Ensuring all gear/kit is secured for sea • Ensuring completion of Bridge Pre-Sail checks • Confirming Stable power before flash-up (MSE) 3) Instructor to lead a guided discussion on a SHOW’s actions in Directing CCR equipment flash up. This can include but is not limited to: • Confirmation of received transmit keys (SATCOM/HF/VHF/FBB) • Liaise with CSE IOT facilitate equipment flash-up • Positional Requirements (Personnel to load crypto, patch circuits, to start-up/shutdown radio equipment and to start-up distribution systems) • Liaise with ORS to receive circuit priorities • Direct comm checks on circuits established • Draft and distribute RADIOSTATE	
Instructor Resources		**References**
1) Markers		1) SSOs

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 002.01.02 – Managing Pre-Sail Checks	
2) Easels 3) 20210826-EO_002.01.02_AIT Activity Floating Flipchart	
Note(s) to Instructor	
Nil.	

EO 002.01 – DIRECT THE FLASH UP AND SHUT DOWN OF COMMUNICATIONS EQUIPMENT
EC 002.01.02 – MANAGE PRE-SAIL CHECKS

EC 002.01.02 – Manage Pre-sail Checks		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.01.02	Classroom	After LG 002.01.02
Time	**Classification**	**Instructor**
50 minutes	Unclassified	MS Nav Comm or above
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct RADHAZ policies and procedures.	
Assessment Steps	**Input:** Trainees will be given a 16-question written assessment that requires them to demonstrate their knowledge with directing the flash-up/shutdown of communications equipment. **Performance –** Trainees will have 50 minutes to complete this assessment. **Output –** Trainee must attain 13/16 to pass this assessment. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20230124-EC_002.01.02(S)		
Note(s) to Instructor		
References and resources may be used for this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 002.02 – DIRECT TRANSMISSION AND RECEPTION INFORMATION BY COMMUNICATIONS CIRCUITS
LG 002.02.01 – DIRECTING TRANSMISSION AND RECEPTION OF INFORMATION BY COMMUNICATION CIRCUITS

LG 002.02.01 – Directing Transmission and Reception of Information by Communication Circuits		
Location Conditions		**Instructional Sequence**
Classified Classroom		After EC 002.01.02
Time	**Classification**	**Instructor**
400 minutes	Classified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Maintain Message Handling Procedures 2) Direct New Day Procedures 3) Supervise Communication Checks • Troubleshoot Military Satellite Connectivity • Troubleshoot Civilian Satellite Connectivity • Monitor Military Satellite Connectivity • Monitor Civilian Satellite Connectivity		1) Brainstorm 2) Guided Discussion 3) Card Sort 4) Best Summary
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct transmission and reception of information by communication circuits.	
Activity Steps	1) Instructor is to conduct an AIT Brainstorm activity with the trainees. Trainees are given the scenario of a ship conducting IMSRTs, conducting various communications exercises, and maintaining comms on a variety of different wavelengths. Trainees are to put themselves in the mindset of a SHOW just starting their watch, using these categories as a guide: Administration, Operation and Personnel. They will be asked to provide examples of what a SHOW must ensure when directing transmission and reception of information by communication circuits. The instructor is to review results with the trainees. 2) Instructor to facilitate Card Sort activity. From the topics listed below, each trainee will choose a subject at random from a hat. Once selected, each trainee will use references provided in order to learn about their chosen topic. After a set time, as dictated by the instructor, each trainee will present their subject to the class. Topics can include but are not limited to: • Adherence to CCR battle rhythm • Ensuring proper CSN numbers on all circuits (S/S, TGOA, MSO) • Ensuring MI log (& Broadcast, S/S, inter-ship files are maintained) (Standardized organization of MI logs and Files) • Routine destruction & INFOSEC 2 • Emergency Destruction • Positional requirements (RATT OP, TG TAC, Broadcast Operator) • Verify frequencies IAW OPTASK COMMS	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 002.02.01 – Directing Transmission and Reception of Information by Communication Circuits	
	• confirming 2-way comms • Monitor chat (Contact with NCS and SOC) • SATCOM DSR sent IAW proper timings • Safe musters are conducted • Staying ahead of the ships Flex • Keeping Bridge teams informed • Conducted full watch turnovers • WGS • FBB 3) Instructor is to have trainees conduct an AIT Best Summary exercise. The trainees are to be divided into two teams (PORT and STBD watch) each team is to prepare a list of what they believe is a SHOWs responsibility. Once each team has their list, a team member is to debrief the opposite team with the contents of their list. Teams are to debrief each other simultaneously. Once both teams have been debriefed, trainees will identify points that the other team has raised that they didn't include and discuss why they have added and their importance to a SHOW.
Instructor Resources	**References**
1) Markers 2) Easel paper	1) CFCD 107 2) SSOs 3) INFOSEC 2
Note(s) to Instructor	
Nil.	

EO 002.02 – DIRECT TRANSMISSION AND RECEPTION INFORMATION BY COMMUNICATIONS CIRCUITS

EC 002.02 – DIRECT TRANSMISSION AND RECEPTION INFORMATION BY COMMUNICATIONS CIRCUITS

EC 002.02 – Direct Transmission and Reception Information by Communication Circuits		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.02.01	Classified Classroom	After LG 002.02.01
Time	**Classification**	**Instructor**
50 minutes	Classified	MS Nav Comm or above
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct transmission and reception of information by communication circuits.	
Assessment Steps	**Input –** Trainees will be given a 10-question written assessment that requires them to demonstrate their knowledge with directing transmission and reception of information by communication circuits. **Performance –** Trainees will each have 50 minutes to complete this assessment. **Output –** Trainee must attain 8/10 SAT to pass this assessment. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20230126-C-NAVCOMM-QL5B-EC_002.02_Direct_Transmission_and_Reception_of_Information_by_Communication_Circuits(I) 2) 20230126-C-NAVCOMM-QL5B-EC_002.02_Direct_Transmission_and_Reception_of_Information_by_Communication_Circuits(S)		
Note(s) to Instructor		
1) Nil.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

EO 002.03 – DIRECT SAR COMMUNICATION PROCEDURES
LG 002.03.01 – MANAGING CCR AND BRIDGE SAR CHECKLIST

LG 002.03.01 – Managing CCR and Bridge SAR Checklist		
Location Conditions	**Instructional Sequence**	
Classroom	After EC 002.02	
Time	**Classification**	**Instructor**
400 minutes	Unclassified	MS or above Nav Comm
Learning Objectives	**Method(s)**	
1) Manage CCR and Bridge SAR Checklists 2) Implement River City Procedures	1) Best Summary 2) Guided Discussion 3) Table Topics	
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct search and rescue communication procedures.	
Activity Steps	1) Instructor is to have trainees conduct AIT activity Best Summary exercise. The trainees are to be divided into two teams (PORT and STBD watch) each team is to prepare a list of what they believe is a SHOWs responsibility directing/initiating search and rescue communications procedures. Once each team has their list, a team member is to debrief the opposite team with the contents of their list. Teams are to debrief each other simultaneously. Once both teams have been debriefed, trainees will identify points that the other team has raised that they didn't include and discuss why they have added and their importance to a SHOW. 2) Instructor will facilitate a guided discussion on trainees' experiences with search and rescue. The instructor is to relate the discussion on the actions a SHOW would take in response to an initiated SAR scenario (ex// MAYDAY alarm on VHF radio) Topics can include but not limited to: • Ready and begin the CCR SAR CHECKLIST • Readying PRCs for dispersal (ensuring batteries are charged and ready, PRCs in holsters) • Positional requirements (Have all on watch personnel available for possible tasking for Boat Comms, Lookouts, personnel on signal lamps and Xenon) • Ensure bridge team has verified EPIRBs and SARTs as well as paddles for the lookouts and tested signal lamps • Ensure TPI on all Emergency Broadcast Circuits • Prepare Iridium phones for use and activation • Prepare to receive circuit priority/alterations from ORS for SAR COMPLAN • River City • PIM considerations • Callsigns when communicating with foreign forces • Iridium phone • Relay Stations • RADHAZ considerations • OPSIGs (Civilian vessels)	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 002.03.01 – Managing CCR and Bridge SAR Checklist	
	3) Trainees are given a copy of a CCR SAR Checklist/Bridge SAR Checklist. 4) Instructor is to review the CCR SAR Checklist/Bridge SAR Checklist with the class, covering all points and discussing each of the trainee's experiences. 5) Instructor is to lead a guided discussion on considerations that must be taken in regard to the River City States during a variety of scenarios while acting as a SHOW in response to these events, including: • Rescue Stations • SAR • Action Stations • Emergency Stations 6) Instructor will facilitate AIT Activity Table Topics. One at a time, trainees will pick from a collection of pre-written talking points related to the content covered in class and speak on that topic (unrehearsed) for an amount of time pre-determined by the instructor. The instructor will debrief the activity once everyone has spoken.
Instructor Resources	**References**
1) Markers 2) Easel paper 3) 20180825-EO_002.03_NAV COMM SAR CHECKLIST 4) 020210719-EO_002.03_Table_Topics .docx 5) 20180824-EO_002.03_NAV COMM SAR CHECKLIST 6) 20180824-EO_002.03_NAV COMM SAR CHECKLIST	1) CFCD 107 2) CAMSAR
Note(s) to Instructor	
Nil.	

EO 002.03 – DIRECT SAR COMMUNICATIO PROCEDURES
EC 002.03 – DIRECT SAR COMMUNICATION PROCEDURES

EC 002.03 – Direct SAR Communication Procedures		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.03	Classroom	After LG 002.03.01
Time	**Classification**	**Instructor**
50 minutes	Unclassified	MS Nav Comm or above
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct search and rescue communication procedures.	
Assessment Steps	**Input –** Trainees will be given a 10-question written assessment that requires them to demonstrate their knowledge with directing Search and Rescue communication procedures. **Performance –** Trainees will each have 50 minutes to complete this assessment. **Output –** Trainee must attain 8/10 to pass this assessment. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20230125-U-NAVCOMM-QL5B-EC_002.03_Direct_SAR_Communication_Procedures(I) 2) 20230125-U-NAVCOMM-QL5B-EC_002.03_Direct_SAR_Communication_Procedures(S)		
Note(s) to Instructor		
1) Nil.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

PO 002 – DIRECT CCR OPERATIONS
PC 002 – DIRECT CCR OPERATIONS

PC 002 – Direct CCR Operations		
Scope	**Location Conditions**	**Instructional Sequence**
PO 002	Classroom	After EC 002.03
Time	**Classification**	**Instructor**
50 minutes	Unclassified	MS Nav Comm or above
Scenario	As a SHOW onboard HMCS HALIFAX, you will be directing CCR operations.	
Assessment Steps	**Input –** Trainees are given a 4-question scenario based written assessment that requires them to demonstrate their knowledge with directing Communications Control Room operations. **Performance –** Trainees are given 50 minutes to complete this EC. **Output –** Trainees must answer 16/20 questions correctly to pass this EC. **Feedback –** Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20210717-U-NAVCOMM-QL5B-PC_002_Direct_CRR_Operations(I) 2) 20210717-U-NAVCOMM-QL5B-PC_002_Direct_CRR_Operations(S)		
Note(s) to Instructor		
1) Nil.		
Assessment Limitations	Nil.	
Impact of Failure	IAW NPTORD 4500-54.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 003 – DIRECT VISUAL COMMUNICATIONS

1. **Performance**: Direct Visual Communications.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) DWAN Workstation; and
      (3) Printer.

   b. **Denied**: Nil.

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall direct visual communications by:

   a. Conducting duties of Flag Deck Supervisor to include:

      (1) Entering and leaving harbour anchoring/mooring flag I procedures; and

      (2) Replenishment at sea (RAS) Flag Deck procedures.

4. **Training Limitations**: Nil.

EO 003.01 – CONDUCT DUTIES OF FLAG DECK SUPERVISOR
LG 003.01.01 – ENTERING AND LEAVING HARBOUR ANCHORING/MOORING FLAG INDIA PROCEDURES

LG 003.01.01 – Entering and Leaving Harbour Anchoring/Mooring Flag I Procedures		
Location Conditions		**Instructional Sequence**
Classroom		After PC 002
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Entering Harbour 2) Leaving Harbour 3) Flag India Procedures 4) Anchoring 5) Mooring 6) Weighing Anchor		1) Guided Discussion 2) Best Summary
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct Flag Deck operations of entering/leaving harbour, anchoring/mooring/weighing anchor and flag India procedures.	
Activity Steps	1) Instructor to lead a guided discussion drawing from the trainees personal experiences with regards to the following: • Entering/leaving harbour (what flags are needed, saluting points and key markers/milestones) • Flag India procedures (nesting) 2) Instructor is to provide trainees with the following handouts and conduct a review: • 20210326-EO_003.01_FLAG_INDIA_PROCEDURES_HANDOUT • 20210326-EO_003.01.01_ EAST_COAST_CHECK_POINTS_SALUTING_POINTS • 20210326-EO_003.01.01_WEST_COAST_CHECK_POINTS_SALUTING_POINTS.png 3) Instructor is to conduct AIT activity Best Summary. Trainees are to be in groups of 3. Each group will prepare a brief for their flag deck as if they were entering Halifax Harbour, coming alongside jetty NB3, outboard of HMCS KINGSTON. Once their brief is complete, one member from each team is to go to another team and brief them. Teams are to discuss the briefs after completion. 4) Instructor to lead a guided discussion drawing from the trainees' personal experiences with regards to the following: • Anchoring/weighing anchor • Mooring	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 003.01.01 – Entering and Leaving Harbour Anchoring/Mooring Flag I Procedures	
	5) Instructor is to provide trainees with the following handout and trainees are to review ATP I VOL II ART201 (FLAG INDIA & FLAG UNIFORM): • 20210326-EO_003.01.01_FLAG_UNIFORM_PROCEDURES_HANDOUT 6) Instructor is to review material covered in ATP I VOL II ART201 (FLAG INDIA & FLAG UNIFORM): • 20210326-EO_003.01.01_FLAG_UNIFORM_PROCEDURES_HANDOUT 7) Instructor is to divide the class into teams of two trainees. Teams will brief either the anchoring, mooring or weighing anchor procedure to the class as Flag Deck I/C, ensuring all milestones within each evolution are covered and conveyed to the rest of the class.
Instructor Resources	**References**
1) 20210326-EO_003.01.01_FLAG_INDIA_PROCEDURES_HANDOUT 2) 20210326-EO_003.01.01_FLAG_UNIFORM_PROCEDURES_HANDOUT 3) 20210326-EO_003.01.01_EAST_COAST_CHECK_POINTS_SALUTING_POINTS 4) 20210326-EO_003.01.01_WEST_COAST_CHECK_POINTS_SALUTING_POINTS.png 5) CANSEATRAINLANT: UNC_2.0 ENTERING AND LEAVING HARBOR.doc (mil.ca) 6) CANSEATRAINLANT: UNC_14.0 CEREMONIES AND FLAGS.doc (mil.ca)	1) ATP I VOL II 2) Manual of Ceremony – 4-2-8 “Dipping Ensigns” 3) ACP 130
Note(s) to Instructor	
Nil.	

EO 003.01.01 – CONDUCT DUTIES OF FLAG DECK SUPERVISOR
EC 003.01.01 – ENTERING AND LEAVING HARBOUR ANCHORING/MOORING FLAG INDIA PROCEDURES

EC 003.01 – Conduct Duties of Flag Deck Supervisor		
Scope	**Location Conditions**	**Instructional Sequence**
LG 003.01.01	Classroom	After LG 003.01.01
Time	**Classification**	**Instructor**
50 minutes	Unclassified	MS or above Nav Comm
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to direct flag deck operations of entering/leaving harbour, anchoring/mooring/weighing anchor and flag INDIA procedures.	
Assessment Steps	**Input:** Trainees will be given a 10-question written assessment **Performance**: Trainees will have 50 minutes to complete the EC **Output**: Trainees must answer 8/10 questions correctly to pass the assessment **Feedback**: Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20210326-EC_003.01.01(I) 2) 20210326-EC_003.01.01(S)		
Note(s) to Instructor		
References and resources may be used to complete this assessment.		
Assessment Limitations	This is a theory based assessment due to resource limitations.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

EO 003.01 – CONDUCT DUTIES OF FLAG DECK SUPERVISOR
LG 003.01.02 – RAS FLAG DECK PROCEDURES

LG 003.01.02 – RAS Flag Deck Procedures		
Location Conditions		**Instructional Sequence**
Classroom		After EC 003.01.01
Time	**Classification**	**Instructor**
400 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Replenishment at Sea Visual Communications a. Day/Night RAS Procedures; Lights, Shapes & Flags b. CORPEN N Procedures c. EMERGENCY 6 2) Position of Flags		1) Guided Discussion 2) Best Summary 3) Slideshow
Scenario	As a SHOW onboard HMCS HALIFAX, you will be expected to identify RAS policy and procedures to ensure personnel safety, as well as communications between units during these evolutions.	
Activity Steps	1) Instructor to lead a guided discussion drawing from the trainees personal experiences with regards to the following: • RAS milestones • CORPEN N procedures • EMERGENCY 6 procedures (response to as Flag Deck I/C) • Day/night procedures • Visual aids used (flags, lights & shapes) • Communication with Bridge team 2) Instructor is to have trainees review ATP I VOL I articles 702 (CORPEN N) 800 (SPEED L), 303 (EMERG 6), 3101, ATP 16 CH 4 ART 0410, 0411, 0412, 0454, CH 5 ART 0500 and provide each trainee with: • 20210326-EO_003.01.02_SEQUENCE_OF_EVENTS • 20210326-EO_003.01.02_FLAG_DECK_PREP_HANDOUT • 20210326-EO_003.01.02_CORPEN_N_EMERG_6 • 20210326-EO_003.01.02_RAS_BY_LIGHT • 20210326-EO_003.01.02_SPEED_L • 20210519-EO_003.01.02_SPEED_L_2 3) Instructor is to lead a review of material covered in ATP I VOL I articles 702 (CORPEN N) 800 (SPEED L), ATP 16 CH 4 ART 0410, 0411, 0412, 0414, 0450, 0454, and the resources given. 4) Instructor to conduct AIT activity Best Summary. The trainees are to be divided into two teams (PORT and STBD watch) each team is to prepare a RAS brief as they are the Flag Deck IC debriefing their subordinates. Instructor is to provide	

LG 003.01.02 – RAS Flag Deck Procedures	
	each group with a RAS scenario (replenishment/replenishing vessel). Once complete, a team member is to debrief the opposite team. Teams are to debrief each other simultaneously. Once both teams have been debriefed, Trainees will identify points that the other team has raised that they didn't include and discuss why they have added and their importance as Flag Deck ICs.
Instructor Resources	**References**
1) 20210326-EO_003.01.02_SEQUENCE_OF_EVENTS 2) 20210326-EO_003.01.02_Flag_deck_prep_handout 3) 20210326-EO_003.01.02_CORPEN_N_EMERG_6 4) 20210326-EO_003.01.02_RAS_BY_LIGHT 5) 20210326-EO_003.01.02_SPEED_L 6) 20210519-EO_003.01.02_SPEED_L_2 7) 20210825-EO_003.01.02_RAS_SCENARIO(1) 8) 20210825-EO_003.01.02_RAS_SCENARIO(2) 9) 20210825-EO_003.01.02_RAS_SCENARIO(3) 10) 20210825-EO_003.01.02_RAS_SCENARIO(4) 11) 20210825-EO_003.01.02_RAS_SCENARIO(5)	1) ATP I Vol II 2) CANSEATRAINLANT - UNC_RAS AND CORPEN NOVEMBER PROCEDURES.ppt (mil.ca) 3) ATP 16
Note(s) to Instructor	
Nil.	

EO 003.01.02 – CONDUCT DUTIES OF FLAG DECK SUPERVISOR
EC 003.01.02 – RAS FLAG DECK PROCEDURES

EC 003.01 – Conduct Duties of Flag Deck Supervisor		
Scope	**Location Conditions**	**Instructional Sequence**
LG 003.01.02	Classroom	After LG 003.01.02
Time	**Classification**	**Instructor**
50 minutes	Unclassified	MS or above Nav Comm
Scenario	As a SHOW onboard HMCS HALIFAX you will be expected to identify RAS policy and procedures to ensure personnel safety, as well as communications between units during these evolutions.	
Assessment Steps	**Input:** Trainees will be given a 10-question written assessment. **Performance**: Trainees will have 50 minutes to complete the EC. **Output**: Trainees must answer 8/10 questions correctly to pass the assessment. **Feedback**: Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1)20210326-EC_003.01.02(S) 2)20210326-EC_003.01.02(I)		
Note(s) to Instructor		
References and resources may be used to complete this assessment.		
Assessment Limitations	This is a theory-based assessment due to resource limitations.	
Impact of Failure	Instructor shall meet with trainee within 1 training day to assess training deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 003 – DIRECT VISUAL COMMUNICATIONS
PC 003 – DIRECT VISUAL COMMUNICATIONS

PC 003 – Direct Visual Communications		
Scope	**Location Conditions**	**Instructional Sequence**
PO 003	Classroom	After EC 003.01.02
Time	**Classification**	**Instructor**
100 minutes	Unclassified	MS or above instructor
Scenario	As a SHOW onboard HMCS HALIFAX, you will be required to supervise Flag Deck operations.	
Assessment Steps	**Input:** Trainees will be given a 15-question written assessment. **Performance**: Trainees will have 100 minutes to complete the EC. **Output**: Trainees must answer 12/15 questions correctly to pass this assessment. **Feedback**: Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) 20210513-PC 003(I) 2) 20210513-PC 003(S)		
Note(s) to Instructor		
1) References and resources may be used to complete this assessment.		
Assessment Limitations	This is a theory-based assessment due to resource limitations.	
Impact of Failure	IAW NPTORD 4500-54.	

PO 004 – ADMINISTER SERVER VIRTUALIZATION

1. **Performance**: Administer Server Virtualization.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) DWAN Workstation; and
      (3) Printer.

   b. **Denied**: Nil.

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall administer server virtualization by:

   a. Building ship servers to an operational state by:

      (1) Installing EXSi 6.5; and
      (2) Deploying VMware templates and servers by:
         (a) Creating a Virtual Machine Template;
         (b) Exporting an OVF Template;
         (c) Deploying a server from a template; and

   b. Implementing vCenter Software by:

      (1) Installing vCenter;
      (2) Configuring vCenter;
      (3) Creating customization files; and
      (4) Deploying servers from a template

   c. Utilizing datastores by:
      (1) Creating a datastore;
      (2) Extending a datastore; and
      (3) Adding additional space to a Windows Server.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

d. Configuring vSwitches and VMware roles and permissions by:

(1) Creating and configuring vSwitch;
(2) Administering VMware roles and permissions.

e. Creating Domain Controllers and joining the Domain by:

(1) Creating a Domain Controller;
(2) Joining the Domain; and
(3) Creating a Secondary Domain Controller.

4. **Training Limitations**: Nil.

EO 004.01 – BUILD SHIP SERVERS TO AN OPERATIONAL STATE
LG 004.01.01 – INSTALLING ESXI 6.5

LG 004.01.01 – Installing ESXI 6.5		
Location Conditions		**Instructional Sequence**
NCITT		After PC 003
Time	**Classification**	**Instructor**
100 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Installing ESXI 6.5		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to install ESXi software.	
Activity Steps	1) Instructor is to present EO_004.01.01_Introduction_to_Virtualization to the class on the projector. 2) Trainees are to complete EO_004.01.01_Installing_ESXi_6.5.	
Instructor Resources		**References**
1) EO_004.01_Introduction_to_Virtualization 2) EO_004.01.01_Installing_ESXi_6.5		Nil.
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 004.01 – BUILD SHIP SERVERS TO AN OPERATIONAL STATE
LG 004.01.02 – CREATING A VIRTUAL MACHINE TEMPLATE

LG 004.01.02 – Creating a Virtual Machine Template		
Location Conditions		**Instructional Sequence**
NCITT		After LG 004.01.01
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Creating a Virtual Machine Template 2) Export OVF Template 3) Deploy a Server from a Template		1) Guided Discussion 2) Lecture
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to create virtual machine templates.	
Activity Steps	1) Instructor is to present EO_004.01.02_Virtual_machine_Templates_Clones_And_Customization_Files to the class on a projector. 2) Instructor is to have trainee complete EO_004.01.02_Creating_a_Virtual_Machine_Template. 3) Upon completion of lab 02, trainees are then to complete EO_004.01.02_Export_OVF_Template. 4) Upon completion of lab 03, trainees are to complete EO_004.01.02_Deploying_a_Server_from_a_Template.	
Instructor Resources		**References**
1) EO_004.01.02_Virtual_machine _Templates_Clones_And_Customization_Files 2) EO_004.01.02_Creating_a_Virtual_Machine_Template 3) EO_004.01.02_Export_OVF_Template 4) EO_004.01.02_Deploying_a_Server_from_a_Template		Nil.
Note(s) to Instructor		
Nil.		

EO 004.01 – BUILD SHIP SERVERS TO AN OPERATIONAL STATE
EC 004.01 – BUILD SHIP SERVERS TO AN OPERATIONAL STATE

EC 004.01 – Build Ship Servers to an Operational State		
Scope	**Location Conditions**	**Instructional Sequence**
EO 004.01	NCITT	After LG 004.01.02
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to create and deploy servers from a template using VMWare.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to create a virtual machine template, then deploy servers from the created template IOT mimic a ships server infrastructure. **Performance –** Trainees will each have 200 minutes to complete this assessment. **Output –** Trainees must achieve 13/13 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ships environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

EC 004.01 CREATE AND DEPLOY TEMPLATES AND SERVERS USING VMWARE			
Item	SAT	UNSAT	Remarks
1. ShipLAN Server Infrastructure			
File and Print Server created	☐	☐	
Exchange Server created	☐	☐	
Backup Server created	☐	☐	
2. VM Settings Verification			
Select one of the servers to have the following VM settings validated:			
Hard Disk 2 set to “Thin Provisioned”	☐	☐	
Network adapter 1 set to “VMXNET 3”	☐	☐	
Video Card’s Total Video Memory set to “8MB”	☐	☐	
VM set to “synchronize guest time with host”	☐	☐	
VM set to run “Server 2016.iso”	☐	☐	
3. Server Manager Settings Verification			
Select a different server to have the following Server Manager settings Validated:			
Time Zone set (AST(Atlantic Standard Time)/PST (Pacific Standard Time))	☐	☐	
IPv6 Disabled	☐	☐	
Windows Firewall turned off	☐	☐	
Remote Desktop enabled	☐	☐	
Windows Server Backup service installed	☐	☐	

Score: /13

Trainee must achieve 13/13 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 004.01 CREATE AND DEPLOY TEMPLATES AND SERVERS USING VMWARE		
Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 004.02 – IMPLEMENT VCENTER SOFTWARE
LG 004.02.01 – INSTALLING VCENTER

LG 004.02.01 – Installing vCenter		
Location Conditions		**Instructional Sequence**
NCITT		After EC 004.01
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) vCenter Install 2) Configure vCenter 3) Creating Customization Files 4) Deploying Servers from a Template		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to install and configure vCenter.	
Activity Steps	1) Instructor is to present EO_004.02_Intro_to_vCenter_Creating_the_vCenter_Datastores_and_LUNs to the class on a projector. 2) Trainees are to complete EO_004.02.01_VCenter_Install. 3) Upon completion of lab 05, trainees are to complete EO_004.02.01_Configure_VCenter. 4) Upon completion of lab 06, trainees are to complete EO_004.02.01_Creating_Customization_Files. 5) Upon completion of lab 07, trainees are to complete EO_004.02.01_Deploying_Servers_from_a_Template.	
Instructor Resources		**References**
1) EO_004.02_Intro_to_vCenter_Creating_the_vCenter_Datastores_and_LUNs 2) EO_004.02.01_VCenter_Install 3) EO_004.02.01_Configure_VCenter 4) EO_004.02.01_Creating_Customization_Files 5) EO_004.02.01_Deploying_Servers_from_a_Template		Nil.
Note(s) to Instructor		
Nil.		

EO 004.03 – UTILIZE DATASTORES
LG 004.03.01 – UTILIZING DATASTORES

LG 004.03.01 – Utilizing Datastore	
Location Conditions	**Instructional Sequence**
NCITT	After LG 004.02.01

LG 004.03.01 – Utilizing Datastore		
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm

Learning Objectives	**Instructional Method(s)**
1) Creating a Datastore 2) Extending a Datastore 3) Add Additional Space to a Windows Server	1) Lecture 2) Self-Study

Scenario	As an ISA onboard HMCS HALIFAX, you will be required to utilize datastores.
Activity Steps	1) Instructor is to present EO_004.03.01_DataStores_and_LUNs to the class. 2) Trainees are to complete EO_004.03.01_Creating_a_Datastore. 3) Once completed, trainees are to complete U- NAVCOMM-QL5B_EO_004.03.01_ Extending_a_Datastore. 4) Once completed, trainees are to complete U- NAVCOMM-QL5B_EO_004.03.01_Adding_Additional_Space_to_a_Windows_Server

Instructor Resources	**References**
1) EO_004.03.01_DataStores_and_LUNs 2) EO_004.03.01_Creating_a_Datastore 3) U- NAVCOMM-QL5B_EO_004.03.01_ Extending_a_Datastore 4) U- NAVCOMM-QL5B_EO_004.03.01_Adding_Additional_ Space_to_a_Windows_Server	Nil.

Note(s) to Instructor
Nil.

EO 004.04 – CONFIGURE VSWITCHES AND VMWARE ROLES AND PERMISSIONS
LG 004.04.01 – CONFIGURING VSWITCHES AND VMWARE ROLES AND PERMISSIONS

LG 004.04.01 –Configuring vSwitches and VMware roles and permissions		
Location Conditions		**Instructional Sequence**
NCITT		After LG 004.03.01
Time	**Classification**	**Instructor**
250 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) vSwitch Configuration and Creation 2) VMware Roles and Permissions		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure vSwitches and VMWare roles and permission.	
Activity Steps	1) Instructor is to present EO_004.04_Virtual_networks_&_VMWare_vSwitches to the class on a projector. 2) Trainees are then to complete EO_004.04.01_vSwitch_Configuration_and_Creation 3) Instructor is to present EO_004.04_Roles_and_Permissions to the class on a projector. 4) Trainees are to complete EO_004.04_Roles_and_Permissions.	
Instructor Resources		**References**
1) EO_004.04_Virtual_networks_&_VMWare _vSwitches 2) EO_004.04.01_vSwitch_Configuration_and _Creation 3) EO_004.04_Roles_and_Permissions 4) EO_004.04_VMWare Roles and Permissions		Nil.
Note(s) to Instructor		
Nil.		

EO 004.04 – CONFIGURE VSWITCHES AND VMWARE ROLES AND PERMISSIONS
EC 004.04 – CONFIGURE VSWITCHES AND VMWARE ROLES AND PERMISSIONS

EC 004.04 – Configure vSwitches and VMware Roles and Permissions		
Scope	**Location Conditions**	**Instructional Sequence**
EO 004.04	NCITT	After LG 004.04.01

EC 004.04 – Configure vSwitches and VMware Roles and Permissions		
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure vSwitch and VMware roles and permissions.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to create a customization file, successfully configure a vSwitch and configure VMware roles and permissions. **Performance –** Trainees will each have 150 minutes to complete this assessment. **Output –** Trainees must achieve 14/14 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ship's environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

PROTECTED B (when complete)

EC 004.04 CONFIGURE VSWITCHES AND VMWARE ROLES AND PERMISSIONS			
Item	SAT	UNSAT	Remarks
1. vSwitch Configuration			
vSwitch 'Test' created	☐	☐	
vSwitch assigned to "VMNIC 1" network adapter	☐	☐	
vSwitch assigned to "VMNIC 2" network adapter	☐	☐	
"VMNIC 2" set as "Standby Adapter"	☐	☐	
Successful ping between ***	☐	☐	
2. Creating VMWare Roles and Permissions			
vCenter Admin role:			
Local Group "vCenter Admins" created	☐	☐	
vCenter Administrator account assigned to "vCenter Admins"	☐	☐	
vCenter Admin Group assigned custom "vCenter Role"	☐	☐	Role will have full access to all functions/options
Host Admin role:			
Local Group "Host Admins" created	☐	☐	
Host Administrator account assigned to "Host Admins"	☐	☐	
Host Admin Group assigned custom "Host Role"	☐	☐	Role will NOT have modify rights for permissions
VM Viewer role:			
Local Group "VM View" created	☐	☐	
VM Viewer account assigned to "VM View"	☐	☐	
VM View Group assigned custom "VM Read Only Access" Role	☐	☐	Role will have most functions "greyed out"

Score: /14

Trainee must achieve 14/14 SATs to pass this assessment.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 004.04 CONFIGURE VSWITCHES AND VMWARE ROLES AND PERMISSIONS		
Positive Feedback	Improvement Feedback	
Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 004.05 – CREATE DOMAIN CONTROLLERS AND JOIN THE DOMAIN
LG 004.05.01 – CREATING A DOMAIN CONTROLLER

LG 004.05.01 – Creating a Domain Controller		
Location Conditions		**Instructional Sequence**
NCITT		After EC 004.04
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Creating a Domain Controller (DC1) 2) Joining the Domain 3) Creating a Secondary Domain Controller		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to create Domain Controllers and joining Domains.	
Activity Steps	1) Instructor is to present EO_004.05_Intro_to_Active_Directory_Domains to the class on a projector. 2) Trainees are to complete EO_004.05.01_Creating a Domain Controller (DC1). 3) Upon completion, trainees are to complete EO_004.05.01_Joining_The_Domain. 4) Upon completion, trainees are to complete EO_004.05.01_Creating_a_Domain_Controller_(DC1). 5) Upon completion, trainees to complete 20230403-EO_004.05.01_Creating a Secondary Domain Controller.	
Instructor Resources		**References**
1) EO_004.05_Intro_to_Active_Directory_Domains 2) EO_004.05.01_Creating a Domain Controller (DC1) 3) EO_004.05.01_Joining_The_Domain 4) EO_004.05.01_Creating_a_Domain_Controller_(DC1) 5) EO_004.05.01_Creating a Secondary Domain Controller		Nil.
Note(s) to Instructor		
Nil.		

EO 004.05 – CREATE DOMAIN CONTROLLERS AND JOIN THE DOMAIN
EC 004.05 – CREATE DOMAIN CONTROLLERS AND JOIN THE DOMAIN

EC 004.05 – Create Domain Controllers and Join the Domain		
Scope	**Location Conditions**	**Instructional Sequence**
EO 004.05	NCITT	After LG 004.05.01
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to create Domain Controllers and join the Domain.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to create a Domain Controller and join the Domain. **Performance –** Trainees will each have 200 minutes to complete this assessment. **Output –** Trainees must achieve 14/14 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ships environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

EC 004.05 CREATE DOMAIN CONTROLLERS AND JOIN THE DOMAIN			
Item	SAT	UNSAT	Remarks
1. Domain Controller 1			
Domain correctly named xxx.local	☐	☐	xxx to be replaced with ship's 3 letter identifier (i.e. DAU.local)
Confirm the following files and applications:			
NTDS.dit database file	☐	☐	Located at C:\ Windows\ NTDS\NTDS.dit
Active Directory Users and Computers	☐	☐	
Active Directory Domain and Trusts	☐	☐	
Active Directory Sites and Services	☐	☐	
DNS Manager	☐	☐	
Confirm that DC1 is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Domain Controllers OU
2. Domain Controller 2			
Confirm the following files and applications:			
NTDS.dit database file	☐	☐	Located at C:\ Windows\ NTDS\NTDS.dit
Active Directory Users and Computers	☐	☐	
Active Directory Domain and Trusts	☐	☐	
Active Directory Sites and Services	☐	☐	
DNS Manager	☐	☐	
Confirm that DC2 is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Domain Controllers OU
3. Join VM to the new domain			
Confirm that VM is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Computers

Score: /14

Trainee must achieve 14/14 SATs to pass this assessment.

EC 004.05 CREATE DOMAIN CONTROLLERS AND JOIN THE DOMAIN		
Positive Feedback	Improvement Feedback	
Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 004 – ADMINISTER SERVER VIRTUALIZATION

PC 004 – ADMINISTER SERVER VIRTUALIZATION

PC 004 – Administer Server Virtualization		
Scope	**Location Conditions**	**Instructional Sequence**
PO 004	NCITT	After EC 004.05
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to administer server virtualization.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to create a Virtual Machine Template, deploy servers from the created template, create a Domain Controller and join the Domain. They will then be required to create and configure a vSwitch and connect it to their new servers. **Performance –** Trainees will each have 200 minutes to complete this assessment. **Output –** Trainees must achieve 21/21 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ships environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	IAW NPTORD 4500-54.	

PROTECTED B (when complete)

PC 004 ADMINISTER SERVER VIRTUALIZATION			
Item	SAT	UNSAT	Remarks
1. Domain Controllers			
Domain correctly named xxx.local	☐	☐	xxx to be replaced with ship's 3 letter identifier (i.e. DAU.local)
Choose one of the Domain Controllers and confirm the following files and applications:			
NTDS.dit database file	☐	☐	Located at C:\ Windows\ NTDS\NTDS.dit
Active Directory Users and Computers	☐	☐	
Active Directory Domain and Trusts	☐	☐	
Active Directory Sites and Services	☐	☐	
DNS Manager	☐	☐	
Confirm that DC1 is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Domain Controllers OU
Confirm that DC2 is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Domain Controllers OU
2. ShipLAN Server Infrastructure			
File and Print server created	☐	☐	
Confirm that the File and Print Server is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Computers
Exchange Server created	☐	☐	
Confirm that the Exchange server is in the correct OU (xxx.local)	☐	☐	Active Directory Users & Computers -> xxx.local -> Computers
3. Create and Configure vSwitch			
vSwitch 'Test' created	☐	☐	
vSwitch assigned to "VMNIC 1" network adapter	☐	☐	
vSwitch assigned to "VMNIC 2" network adapter	☐	☐	
"VMNIC 2" set as "Standby Adapter"	☐	☐	
Confirm the following servers are assigned to the vSwitch:			
Domain Controller 1	☐	☐	

PC 004 ADMINISTER SERVER VIRTUALIZATION			
Domain Controller 2	☐	☐	
File and Print Server	☐	☐	
Exchange Server	☐	☐	

Score: /21

Trainee must achieve 21/21 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 005 – ADMINISTER SERVER MANAGEMENT

1. **Performance**: Administer Server Management.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) DWAN Workstation; and
      (3) Printer.

   b. **Denied**: Nil.

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall administer server management by:

   a. Administering the DNS server role by:
      (1) Adding the DNS server role; and
      (2) Adding resource records for XXX.Local.

   b. Managing the printer service by:

      (1) Installing the print service.

   c. Creating scripts and profiles by:

      (1) Enabling logon scripts; and
      (2) Creating roaming profiles.

4. **Training Limitations**: Nil.

EO 005.01 – ADMINISTER DNS SERVER ROLE
LG 005.01.01 – ADDING THE DNS SERVER ROLE

LG 005.01.01 – Adding the DNS Server Role		
Location Conditions		**Instructional Sequence**
NCITT		After PC 004
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Adding the DNS Server Role 2) Adding Resource Records for XXX.Local		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to add the DNS server role.	
Activity Steps	1) Instructor is to present EO_005.01_Installing_and_Configuring_Server_Role-DNS to the class on the projector. 2) Trainees are to complete EO_005.01_Adding the DNS Server Role. 3) Once complete, trainees are to complete EO_005.01.01_ Adding DNS Resource Records.	
Instructor Resources		**References**
1) EO_005.01_Installing_and_Configuring_Server_Role-DNS 2) EO_005.01_Adding the DNS Server Role 3) EO_005.01.01_ Adding DNS Resource Records		Nil.
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 005.01 – ADD THE DNS SERVER ROLE
EC 005.01 – ADD THE DNS SERVER ROLE

EC 005.01 – Adding the DNS Server Role		
Scope	**Location Conditions**	**Instructional Sequence**
EC 005.01	NCITT	After LG 005.01.01
Time	**Classification**	**Instructor**
100 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to add the DNS server role.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to add the DNS server role. **Performance –** Trainees will each have 100 minutes to complete this assessment. **Output –** Trainees must achieve 6/6 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ships environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

EC 005.01 ADDING THE DNS SERVER ROLE			
Item	SAT	UNSAT	Remarks
Add the DNS Server Role			
PART 1 - Secondary zone created (XXX.local on PV10001)	☐	☐	
PART 2 - Enable zone transfers (Between CV10001 and PV10001)	☐	☐	
PART 3 - New NS record appears with in details pane of DNS Manager (ESQ/HFX-XXX-PV10001.XXX.local)	☐	☐	
PART 4 – PV10001 server found within newly-created zone (Reverse Lookup Zone)	☐	☐	
PART 5 – New zone appears in DNS Manager	☐	☐	

Score: /6

Trainee must achieve 6/6 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 005.02 – MANAGE THE PRINTER SERVICE
LG 005.02.01 – INSTALLING THE PRINT SERVICE

LG 005.02.01 – Installing the Print Service		
Location Conditions		**Instructional Sequence**
NCITT		After EC 005.01
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Installing the Print Service 2) Installing Printers		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to manage the printer service.	
Activity Steps	1) Instructor to present EO_005.02_Managing_Print_Servers to the class on the projector. 2) Trainees are to complete EO_005.02.01_Installing_the_Print_Service.	
Instructor Resources		**References**
1) EO_005.02_Managing_Print_Servers 2) EO_005.02.01_Installing_the_Print_Service		Nil.
Note(s) to Instructor		
Nil.		

EO 005.02 – MANAGE THE PRINTER SERVICE
EC 005.02 – MANAGE THE PRINTER SERVICE

EC 005.02 – Managing the Printer Service		
Scope	**Location Conditions**	**Instructional Sequence**
EO 005.02	NCITT	After LG 005.02.01
Time	**Classification**	**Instructor**
100 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to manage the printer service.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to install the print service and install printers. **Performance –** Trainees will each have 100 minutes to complete this assessment. **Output –** Trainees must achieve 3/3 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ships environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment. 3) In order to ensure trainees have successfully the print service, trainees are to print a test page upon completion.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

EC 005.02 MANAGING THE PRINTER SERVICE			
Item	SAT	UNSAT	Remarks
Managing The Printer Service			
Installed the 'Print and Document Services' server role	☐	☐	
Confirmation of install with 'Print Management Window' appearing with no errors	☐	☐	
Confirmation of entries for 'Print Services and 'Roles and Server Groups'	☐	☐	

Score: /3

The trainee must achieve 3/3 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 005.03 – CREATE SCRIPTS AND PROFILES
LG 005.03.01 – CREATING SCRIPTS AND PROFILES

LG 005.03.01 – CREATING SCRIPTS AND PROFILES		
Location Conditions		**Instructional Sequence**
NCITT		After EC 005.02
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Enabling Logon Scripts 2) Creating Roaming Profiles		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to create scripts and profiles.	
Activity Steps	1) Instructor is to present EO_005.03_Scripting to the class on the projector. 2) Trainees are to complete EO_005.03.01_Logon_Scripts. 3) Once completed, they are to complete U-NAVCOMM-QL5B-EO_005.03_Roaming Profiles.	
Instructor Resources		**References**
1) EO_005.03_Scripting 2) EO_005.03.01_Logon_Scripts 3) U-NAVCOMM-QL5B-EO_005.03_Roaming Profiles		Nil.
Note(s) to Instructor		
Nil.		

EO 005.03 – CREATING SCRIPTS AND PROFILES
EC 005.03 – CREATE SCRIPTS AND PROFILES

EC 005.03 – Creating Scripts and Profiles		
Scope	**Location Conditions**	**Instructional Sequence**
EO 005.03	NCITT	After LG 005.03.01
Time	**Classification**	**Instructor**
100 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to create scripts and profiles.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to create scripts and roaming profiles **Performance –** Trainees will each have 100 minutes to complete this assessment. **Output –** Trainees must achieve 5/5 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ship's environment within vCenter is ready prior to starting this EC. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

EC 005.03 CREATING SCRIPTS AND PROFILES			
Item	SAT	UNSAT	Remarks
1. Create Logon Scripts			
Drive mapping successful on CV10001 (logged in as XXXAdmin)	☐	☐	
Drive mapping successful on PV10001 (logged in as XXXAdmin)	☐	☐	
2. Creating a Roaming Profile			
Q: drive created on VWS10001	☐	☐	
.V6 folder inside Admins Q: drive (PV10001)	☐	☐	

Score: /4

Trainee must achieve 4/4 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 005 – ADMINISTER SERVER MANAGEMENT
PC 005 – ADMINISTER SERVER MANAGEMENT

PC 005 – Administer Server Management		
Scope	**Location Conditions**	**Instructional Sequence**
PO 005	NCITT	After EC 005.03
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to administer server management.	
Assessment Steps	**Input –** Trainees will be given a NCITT workstation with VMware and vCenter already enabled. Each trainee will then be required to install the print service, enable login scripts and create roaming profiles. **Performance –** Trainees will each have 200 minutes to complete this assessment. **Output –** Trainees must achieve 7/7 SATs to pass this assessment. **Feedback –** Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) Instructor Lab Binder		
Note(s) to Instructor		
1) Ensure each ships environment within vCenter is ready prior to starting this EC. 2) In order to ensure trainees have successfully the print service, trainees are to print a test page upon completion. 3) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	IAW NPTORD 4500-54.	

PROTECTED B (when complete)

PC 005 ADMINISTER SERVER MANAGEMENT			
Item	SAT	UNSAT	Remarks
1. Managing the Printer Service			
Installed the 'Print and Document Services' Server Role	☐	☐	
Confirmation of install with 'Print Management Window' appearing with no errors	☐	☐	
Confirmation of entries for 'Print Services and 'Roles and Server Groups'	☐	☐	
2. Create Login Scripts			
Drive mapping successful on CV10001 (logged in as XXXAdmin)	☐	☐	
Drive mapping successful on PV10001 (logged in as XXXAdmin)	☐	☐	
3. Creating a Roaming Profile			
Q: drive created on VWS10001	☐	☐	
.V6 folder inside Admins Q: drive (PV10001)	☐	☐	

Score: /7

Trainee must achieve 7/7 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 006 – ADMINISTER MICROSOFT EXCHANGE SERVER

1. **Performance**: Administer Microsoft Exchange Server.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) DWAN Workstation; and
      (3) Printer.

   b. **Denied**: Nil.

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall administer Microsoft Exchange server by:

   a. Maintaining the Exchange Server by:

      (1) Configuring Exchange Control Panel (ECP);

      (2) Configuring Mailboxes to include;
         (a) Recipient Mailboxes;
         (b) Positional Mailboxes; and
         (c) Group Mailboxes; and
         (d) Manage Network User Accounts (ET0320)

      (3) Configuring Client Access and Outlook Web Access (OWA);

      (4) Configuring Transport Rules;

      (5) Configuring Groups and Roles to include:
         (a) Security;
         (b) Distribution; and
         (c) Dynamic Groups

      (6) Installing the Exchange Management Tool.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

4. **Training Limitations**: Nil.

EO 006.01 – MAINTAIN THE EXCHANGE SERVER
LG 006.01.01 – CONFIGURING ECP

LG 006.01.01 – Configuring ECP		
Location Conditions		**Instructional Sequence**
NCITT		After PC 005
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Instructional Method(s)**
1) Configure ECP		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure ECP.	
Activity Steps	1) Instructor is to present EO_006.01_Introduction_to_Exchange to the class on the projector. Instructor is to emphasize the difference between EMS and EMC to the class. 2) Instructor is to present EO_006.01_Exchange_Recipient_Objects to the class on the projector. 3) Trainees are to complete EO_006.01_Configuring Exchange Administrative Center once the prior presentation has concluded.	
Instructor Resources		**References**
1) EO_006.01_Introduction_to_Exchange 2) EO_006.01_Exchange_Recipient_Objects 3) EO_006.01_Configuring Exchange Administrative Center		Nil.
Note(s) to Instructor		
Nil.		

EO 006.01 – MAINTAINING THE EXCHANGE SERVER
LG 006.01.02 – CONFIGURING MAILBOXES

LG 006.01.02 – Configuring Mailboxes		
Location Conditions		**Instructional Sequence**
NCITT		After LG 006.01.01
Time	**Classification**	**Instructor**
250 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Configure Mailboxes to Include; a. Recipient Mailboxes; b. Positional Mailboxes; and c. Group Mailboxes.		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure mailboxes within the Exchange server.	
Activity Steps	1) Instructor is to display 20210824-EO_006.01.02_Mailbox_Config to the class on a projector. 2) Trainees are to complete 20230406-U-NAVCOMM-QL5B-EO_006.01.02_Mailboxes once the prior presentation has concluded.	
Instructor Resources		**References**
1) EO_006.01.02_Mailbox_Config 2) U-NAVCOMM-QL5B-EO_006.01.02_Mailboxes		Nil.
Note(s) to Instructor		
Nil.		

EO 006.01 – MAINTAINING THE EXCHANGE SERVER
LG 006.01.03 – CONFIGURING CLIENT ACCESS AND OWA

LG 006.01.03 – Configuring Client Access and OWA		
Location Conditions		**Instructional Sequence**
NCITT		After LG 006.01.02
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Configure Client Access 2) Configure OWA		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure client access and OWA within Exchange.	
Activity Steps	1) Instructor is to present EO_006.01.03_Configure_Client_Access and EO_006.01.03_OWA to the class on a projector. 2) Trainees are to EO_006.01.03_ Exchange Client Access once the prior presentations are complete.	
Instructor Resources		**References**
1) EO_006.01.03_Configure_Client_Access 2) EO_006.01.03_OWA 3) EO_006.01.03_ Exchange Client Access		Nil.
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 006.01 – MAINTAINING THE EXCHANGE SERVER
LG 006.01.04 – CONFIGURING TRANSPORT RULES

LG 006.01.04 – Configuring Transport Rules		
Location Conditions		**Instructional Sequence**
NCITT		After LG 006.01.03
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Configure Transport Rules		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure transport rules within Exchange	
Activity Steps	1) Instructor is to present EO_006.01.04_Message_Transport to the class on a projector. 2) Trainees are to complete EO_006.01.04_Exchange Transport Rules once the prior presentation has completed.	
Instructor Resources		**References**
1) EO_006.01.04_Message_Transport 2) EO_006.01.04_Exchange Transport Rules		Nil.
Note(s) to Instructor		
Nil.		

EO 006.01 – MAINTAINING THE EXCHANGE SERVER
LG 006.01.05 – CONFIGURING GROUPS, ADDRESS LIST AND RBAC

LG 006.01.05 – Configuring Groups, Address List and RBAC		
Location Conditions	**Instructional Sequence**	
NCITT	After LG 006.01.04	
Time	**Classification**	**Instructor**
250 minutes	Unclassified	MS or above Nav Comm
Learning Objectives	**Method(s)**	
1) Configure Groups to Include: a. Security Groups; b. Distribution Groups; and c. Dynamic Groups	1) Lecture 2) Self-Study	
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to configure groups, address lists and RBAC within Exchange.	
Activity Steps	1) Instructor is to present EO_006.01.05_Configuring_Groups_Address_Lists_and_RBAC to the class on the projector. 2) Trainees are to complete EO_006.01.05_Exchange_Groups_and_Roles once the prior presentation has concluded.	
Instructor Resources	**References**	
1) EO_006.01.05_Configuring_Groups_Address_Lists_and_RBAC 2) EO_006.01.05_Exchange_Groups_and_Roles	Nil.	
Note(s) to Instructor		
Nil.		

EO 006.01 – MAINTAINING THE EXCHANGE SERVER
LG 006.01.06 – INSTALLING THE EXCHANGE MANAGEMENT TOOL ON A WORKSTATION

LG 006.01.06 – Installing the Exchange Management Tool on a Workstation		
Location Conditions		**Instructional Sequence**
NCITT		After LG 006.01.05
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Learning Objectives		**Method(s)**
1) Install the Exchange Management Tool		1) Lecture 2) Self-Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to Install the Exchange Management Tool.	
Activity Steps	1) Instructor is to present EO_006.01.06_Installing_Exchange_Management_Tool to the class on the projector. 2) Trainees are to complete EO_006.01.06_Exchange_Management_Tool once the prior presentation has concluded.	
Instructor Resources		**References**
1) EO_006.01.06_Installing_Exchange 2) EO_006.01.06_Exchange_Management_Tools		Nil.
Note(s) to Instructor		
Nil.		

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 006.01 – MAINTAIN THE EXCHANGE SERVER
EC 006.01 – MAINTAIN THE EXCHANGE SERVER

EC 006.01 – Maintain the Exchange Server		
Scope	**Location Conditions**	**Instructional Sequence**
EO 006.01	NCITT	After LG 006.01.06
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to maintain the Exchange server.	
Assessment Steps	**Input** – Trainees will be given a NCITT workstation with Microsoft Exchange Tools already enabled. Each trainee will then be required to create exchange mailboxes, create rules and positional accounts. **Performance** – Trainees will have 150 minutes to complete this assessment. **Output** – Trainees must attain 5/5 SAT to pass this assessment. **Feedback** – Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) EC_006.01.docx		
Note(s) to Instructor		
1) Microsoft Exchange tools to be installed prior to commencement; direct verification to NCITT maintainers to ensure this environment is prepared prior to PC commencement. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

EC 006.01 Maintaining the Exchange Server			
Item	SAT	UNSAT	Remarks
Global Distribution Group created	☐	☐	
Account creation	☐	☐	
Positional account accessed through personal account	☐	☐	
Transport Rule created & applied	☐	☐	
Address list created with limited accessibility	☐	☐	

Score: /5

Trainee must achieve 5/5 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 006 – ADMINISTER MICROSOFT EXCHANGE
PC 006 – ADMINISTER MICROSOFT EXCHANGE

PC 001 – Administer Microsoft Exchange		
Scope	**Location Conditions**	**Instructional Sequence**
PO 006	NCITT	After EC 006.01
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS or above Nav Comm
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to administer Microsoft Exchange	
Assessment Steps	**Input** – Trainees will be given a NCITT workstation with Microsoft Exchange Tools already enabled. Each trainee will then be required to create Exchange mailboxes, create rules and positional accounts. **Performance** – Trainees will have 150 minutes to complete this assessment. **Output** – Trainees must attain 5/5 SATs to pass this assessment. **Feedback** – Trainees will receive feedback from instructional staff on or before the next training day.	
Instructor Resources		
1) PC_006		
Note(s) to Instructor		
1) Instructor to ensure email sent as +ISA from its positional account 2) Microsoft Exchange Tools to be installed prior to commencement; direct verification to NCITT maintainers to ensure this environment is prepared prior to PC commencement. 3) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	IAW NPTORD 4500-54.	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PROTECTED B (when complete)

PC 006 Administer Microsoft Exchange			
Item	SAT	UNSAT	Remarks
Account creation	☐	☐	
+ISA account accessed through trainee account	☐	☐	
Transport Rule created and implemented	☐	☐	
Email sent & received by all addressees	☐	☐	
Address List created with limited access	☐	☐	

Score: /5

The trainee must achieve 5/5 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 007 – ADMINISTER INFORMATION SYSTEMS (IS)

1. **Performance**: Administer Information Systems.

2. **Conditions**:

   a. **Given**:

      (1) References;
      (2) DWAN Workstation; and
      (3) Printer.

   b. **Denied**: Nil.

   c. **Environment**:

      (1) Alongside or at sea;
      (2) Any weather conditions; and
      (3) Day or night.

3. **Standard**: The CISN Administrator shall Administer Information Systems by:

   a. Administering IS Troubleshooting Procedures by:

      (1) Supervising Rectification of Workstation/Server Faults (ET0420);
      (2) Supervising Troubleshooting of Workstation/Server Faults (ET0400);
      (3) Managing Network Monitoring Tools;
      (4) Logging Incident and Metrics of Alerts (ET0380);
      (5) Supervising Liaison with Local IS Authorities; and
      (6) Responding to Helpdesk Calls (ET0330).

   b. Administering IS Equipment by:

      (1) Supervising Installation of Software on a Workstation/Server (ET0150);
      (2) Managing Installation of Software on a Workstation/Server (ET0160);
      (3) Supervising Installation of Hardware on Workstation/Server (ET0200);
      (4) Managing Installation of Hardware on Workstation/Server (ET0210);
      (5) Preparing IS Status Briefs (ET0570); and
      (6) Providing Technical Overview of IS Equipment.

4. **Training Limitations**: Nil.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EO 007.01 – ADMINISTER IS TROUBLESHOOTING PROCEDURES
LG 007.01.01 – CONDUCTING FAULT FINDING

LG 007.01.01 – Conducting Fault Finding		
Location Conditions		**Instructional Sequence**
NCITT		After PC 006
Time	**Classification**	**Instructor**
500 minutes	Unclassified	MS Nav Comm or above
Learning Objectives		**Instructional Method(s)**
1) Supervise Rectification of workstation/Server faults; 2) Supervise Troubleshooting of Workstation Faults (ET0400); 3) Supervise Rectification of Workstation Faults (ET0420); and 4) Manage Network Monitoring Tools		1) Guided Discussion 2) T-Chart Activity 3) Presentation 4) Collaborative Problem Solving 5) Scenario Based Learning Activity
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to conduct fault finding.	
Activity Steps	1) Instructor is to conduct a guided discussion, engaging trainees' prior experience in IS troubleshooting. The following topics shall be discussed: a) Fault Monitoring on workstations and servers b) Event Viewer c) Task Manager d) Help Menu/File e) Server Diagnostic Tools f) Report Logging g) Re-imaging h) Approved Software i) Windows/App/Software Updates j) Antivirus k) Command Prompt Basics l) Google Search m) Simple Solutions to Common Problems n) NavIS SOP 2) Instructor to facilitate T-Chart Activity. Trainees are to be split into groups of two or three. Each group is to list methods and skills that are used in order to troubleshoot errors with particular focus on Information Technology. At the end of the activity, a member from each group is to present their group's findings to the class. Instructor may reference EO_007.01_TChart_Activity_Instructor. 3) Instructor to present EO_007.01_Conduct_Fault_Finding 4) Instructor to facilitate Collaborative Problem-Solving Activity. Instructor to display EO_007.01_Collaborative_Problem_Solving_Activity_1.doc. Trainees to	

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 007.01.01 – Conducting Fault Finding	
be divided into groups of two or three. Each group is to prepare a solution. Upon completion, each team will present their solution to the class for further discussion. These solutions shall include methods of troubleshooting and resources to correct the presented scenario. Instructor to debrief activity. 5) Instructor to facilitate a Scenario Based Learning Activity. Using the tools learned previously, they will be given the scenario handout EO_007.01.01_Scenario_HandOut. Using this document, they are to identify the troubleshooting tools they would use to diagnose the issue in the scenario. Instructor to debrief the activity.	
Instructor Resources	**References**
1) EO_007.01_Guided_Discussion_Activity_Instructor 2) EO_007.01_TChart_Activity_Instructor 3) EO_007.01_Conduct_Fault_Finding 4) EO_007.01_Collaboractive_Problem_Solving_Activity_1 5) EO_007.01.01_Scenario_HandOut	1) Information Technology (IT) Certifications & Training | CompTIA 2) NavIS (mil.ca) 3) BIS Ship Support / Fleet ISA - Links - All Links (mil.ca) 4) BIS - Services (mil.ca) 5) Base Information Services - Home (mil.ca)
Note(s) to Instructor	
1) Instructors are recommended to have guest speakers (i.e. Fleet ISA & Ship Support Staff).	

EO 007.01 – ADMINISTER IS TROUBLESHOOTING PROCEDURES
LG 007.01.02 – REGULATING IS TROUBLE TICKET

LG 007.01.02 – Regulating IS Trouble Ticket		
Location Conditions		**Instructional Sequence**
NCITT		After LG 007.01.01
Time	**Classification**	**Instructor**
500 minutes	Unclassified	MS Nav Comm or above
Learning Objectives		**Instructional Method(s)**
1) Supervise liaison with IS local authorities 2) Respond to Helpdesk Calls (ET0330) 3) Log incident and metrics of alerts (ET0380)		1) Guided Discussion 2) Power Point Presentation 3) Scenario Based Learning Activity
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to Regulate Information Systems Trouble Ticket.	
Activity Steps	1) Instructor is to conduct a guided discussion on Assyst Trouble tickets. During this discussion, instructor is to pull the 5Ws from trainees (who, what, when, where and why) do we use trouble tickets. a) Using EO_007.01.02_TroubleTicket_Examples, the instructor is to demonstrate complete and incomplete examples of trouble tickets. Instructor to discuss below: • What is a trouble ticket? • EITSM • Types of services • Elevating a trouble ticket • Logging a trouble ticket while at sea • After-hour support • Asset search b) Instructor to confirm the 5Ws with the below information: • Network that is affected by the issue (Example: DWAN, CSNI, etc.) • Summarize the service/equipment that is impacted (Example: Mail flow on the ShipLAN Microsoft Exchange Server) • Number of users affected identified (Is it happening to everyone or only a few users?) • Initial Investigation (How was the issue discovered?) • Troubleshooting Steps Taken (What have you tried so far to resolve the issue?) • Active Alarms (Example: Amber lights, error pop-ups, etc.) • Documentation Utilized (Various NavIS SOPs include brief troubleshooting sections) 3) Instructor to present EO_007.01.02_Submitting_Assyst_Tickets	

LG 007.01.02 – Regulating IS Trouble Ticket	
4) Instructor to facilitate Scenario Based Learning Activity. Trainees to be divided into groups of two or three. Each group will be given a scenario from EO_007.01.02_SBL_Scenarios. Trainees to be provided trouble ticket template. Each group will fill in the required information needed to complete the trouble ticket. At the end of the activity, each group will have a member present their results. Instructor to debrief activity.	
Instructor Resources	**References**
1) EO_007.01.02_TroubleTicket_Examples 2) EO_007.01.02-Submitting_Assyst_Ticket 3) EO_007.01.02_SBL_Scenarios	1) Information Technology (IT) Certifications & Training | CompTIA 2) NavIS (mil.ca) 3) BIS Ship Support / Fleet ISA - Links - All Links (mil.ca) 4) BIS - Services (mil.ca) 5) Base Information Services - Home (mil.ca) 6) Https://eitsm-gstie.forces.mil.ca/web/hfx/
Note(s) to Instructor	
2) Instructors are recommended to have guest speakers when possible (i.e. Fleet ISA, Ship Support Staff, Help Desk etc.).	

EO 007.01 – ADMINISTER IS TROUBLESHOOTING

EC 007.01 – ADMINISTER IS TROUBLESHOOTING PROCEDURES

EC 007.01 – Administer IS Troubleshooting Procedures		
Scope	**Location Conditions**	**Instructional Sequence**
EO 007.01	NCITT	After LG 007.01.02
Time	**Classification**	**Instructor**
150 minutes	Unclassified	MS Nav Comm or above
Scenario	As the ISA onboard HMCS HALIFAX, you will be required to administer IS troubleshooting procedures	
Assessment Steps	**Input***:* Trainees will be given 2-scenarios that requires them to demonstrate their knowledge to administer IS troubleshooting procedures. **Performance**: Trainees will have 150 minutes to complete this EC. **Output***:* Trainees must achieve 14/14 SATs to pass this assessment. **Feedback**: Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) EC_007.01_Formatted_Assyst_Ticket		
Note(s) to Instructor		
1) Instructor shall meet with trainee within 1 training day to assess training deficiencies. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Retest if necessary.	

PROTECTED B (when complete)

EC 007.01 Administer IS Troubleshooting Procedures			
Item	SAT	UNSAT	Remarks
1. Scenario #1 Trouble Ticket Creation			
Trouble ticket created to include:			
Specified network affected	☐	☐	
Possible root cause identified	☐	☐	
Listed number of users affected	☐	☐	
Described initial investigation	☐	☐	
Described troubleshooting steps taken	☐	☐	
Included active alarms and errors	☐	☐	
Specified documentation utilized	☐	☐	If applicable
2. Scenario #2 Trouble Ticket Creation			
Trouble ticket created to include:			
Specified network affected	☐	☐	
Possible root cause identified	☐	☐	
Listed number of users affected	☐	☐	
Described initial investigation	☐	☐	
Described troubleshooting steps taken	☐	☐	
Included active alarms and errors	☐	☐	
Specified documentation utilized	☐	☐	If applicable

Score: /14

Trainee must achieve 10/14 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard

Assessor
I have discussed the results of this assessment with the trainee.

Name:	Date:	Signature:

Trainee
I have been debriefed on the result of this assessment.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 007.01 Administer IS Troubleshooting Procedures		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 007.02 – ADMINISTER IS EQUIPMENT
LG 007.02.01 – MANAGING WORKSTATION AND SERVER INFRASTRUCTURE

LG 007.02.01 – Managing Workstation and Server Infrastructure		
Location Conditions		**Instructional Sequence**
NCITT		After EC 007.01
Time	**Classification**	**Instructor**
850 minutes	Unclassified	MS Nav Comm or above
Learning Objectives		**Instructional Method(s)**
1) Supervise Installation of Software on a Workstation/Server (ET0150) 2) Manage Installation of Software on a Workstation/Server (ET0160) 3) Supervise Installation of Hardware on Workstation/Server (ET0200) 4) Manage Installation of Hardware on Workstation/Server (ET0210)		1) Guided Discussion 2) Power Point Presentation 3) Case Study
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to manage workstation and server infrastructure.	
Activity Steps	1) Instructor is to facilitate a guided discussion using 20230201-EO_007.02.01_Guided_Discussion.doc to gauge trainees' experience with managing workstation and servers. The following topics shall be discussed: a) User profile management b) Data Storage management c) Bandwidth Management d) Backups e) Use of Approved Software f) Use of Secondary and Tertiary IS Equipment g) Acquiring updated network images h) File Access and Permissions 2) Instructor to present EO_007.02.01_Manage_Workstation_and_Server_Infrastructure 3) Instructor to facilitate a Case Study activity. Trainees to be given EO_007.02.01_Case_Study_Activity.doc. Instructor to divide the class into groups of two. Each group will come up with a solution and determine the course of action based on the given scenario. At the end of the activity each group will have a member present their answer. Instructor to debrief activity. 4) Instructor to conduct walk-through activity. Instructor to use EO_007.02.01_McAfee_E-Policy_Orchestrator in order to walk trainees through McAfee usage.	

LG 007.02.01 – Managing Workstation and Server Infrastructure	
	5) Trainees are to complete EO_007.02.01_Creating_Groups_and_Organizational_Units 6) Trainees are to complete EO_007.02.01_Managing_Access_to_Resources 7) Instructor is to then facilitate Scenario Based Learning activities by using EO_007.02.01_Guided_Troubleshooting_Activities(I).
Instructor Resources	**References**
1) EO_007.02.01_Guided_Discussion 2) EO_007.02.01_Manage_Workstation_and_Server_Infrastructure 3) EO_007.02.01_Case_Study_Activity 4) EO_007.02.01_McAfee_E-Policy_Orchestrator 5) EO_007.02.01_Creating_Groups_and_Organizational_Units 6) EO_007.02.01_Managing_Access_to_Resources 7) EO_007.02.01_Guided_Troubleshooting_Activities(I)	1) Information Technology (IT) Certifications & Training | CompTIA 2) NavIS (mil.ca) 3) BIS Ship Support / Fleet ISA - Links - All Links (mil.ca) 4) BIS - Services (mil.ca) 5) Base Information Services - Home (mil.ca)
Note(s) to Instructor	
Nil.	

EO 007.02 – ADMINISTER IS EQUIPMENT
LG 007.02.02 – CONDUCTING IS STATUS BRIEF

LG 007.02.02 – Conducting Information Systems Status Brief		
Location Conditions		**Instructional Sequence**
NCITT		After LG 007.02.01
Time	**Classification**	**Instructor**
850 minutes	Unclassified	MS Nav Comm or above
Learning Objectives		**Instructional Method(s)**
1) Preparing IS Status Briefs (ET0570) 2) Providing Technical Overview of IS Equipment 3) Report Initial IS Incident 4) Prepare IS Status Brief		1) Think-Pair-Share 2) Guided Discussion 3) Case study 4) Fish Bowl 5) Three Stars
Scenario	As an ISA onboard HMCS HALIFAX, you will conduct Information Systems Status Brief.	
Activity Steps	1) Instructor is to have trainees conduct AIT activity Think-Pair-Share exercise. The trainees are to be given the scenario of an IS incident. Trainees are to individually come up with a list of steps they would take to respond to the incident, then how they would report the incident. After 5 minutes, they are then to pair up with another trainee – lists may be adjusted. After another 5 minutes, trainees are to share their findings with the class. Instructor to then debrief the activity. 2) Instructor to lead guided discussion on reporting IS incidents. Instructor to draw from personal experience with IS incidents as well as have the trainees share their experiences. 3) Instructor to facilitate Case Study activity. Instructor to provide trainees with EO_007.07.02_Case_Study. With this Case Study, as a class, the instructor is to discuss how the issue is detected, reported, and rectified with the CoC. Discussion is to reference future mitigation, continuing operations and rectification of networks / services. • What networks / services are now unavailable? • Impact to ships' operations • Who to report incident to? 4) Instructor to lead guided discussion on an IS status brief. Instructor to display Command Update Brief (CUB) slide and explain the IS information presented. 5) Instructor to facilitate Fishbowl activity. Each trainee is to be given the information that is input on their slide using EO_007.02.02_Briefing_Slides_Scenarios. Once completed the class will be	

LG 007.02.02 – Conducting Information Systems Status Brief	
divided into two teams (PORT and STBD). Each trainee will have the opportunity to be the ISA presenting their slide to their teams. Trainees are to provide the status of each network, identify applicable issues, provide troubleshooting actions taken, give recommendations and provide a rectification time estimate. 6) Instructor is to have trainees conduct AIT activity Three Stars/Think-Pair-Share exercise. Each trainee is to decide what they think are the three most important takeaways from the class. The trainees will then find a partner and decide as a team what their three stars are from the lesson. Once completed, the instructor will lead the class and decide what the class's top three takeaways are from the lesson. Instructor to debrief the activity.	
Instructor Resources	**References**
1) EO_007.02.02_Case_Study_Activity_Scenario_Instructor_Aid 2) EO_007.02.02_Briefing_Slides 3) EO_007.02.02_Briefing_Slides_Scenarios 4) EO_007.02.02_Think_Pair_Share_Scenario 5) EO_007.02.02_Case_Study	1) NavIS Home (mil.ca) – Naval Information Systems
Note(s) to Instructor	
1) Instructor to ensure ISAs are aware of engaging their CoC regarding IS incidents. 2) Instructors to show trainees NavIS resources (NavIS Home (mil.ca) – Navy Information Systems). SOPs and Links tabs are to be shown.	

EO 007.02 – ADMINISTER IS EQUIPMENT
EC 007.02 – ADMINISTER IS EQUIPMENT

EC 007.02 – Administer IS Equipment		
Scope	**Location Conditions**	**Instructional Sequence**
EO 007.02	NCITT	After LG 007.02.02
Time	**Classification**	**Instructor**

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 007.02 – Administer IS Equipment		
150 minutes	Unclassified	MS Nav Comm or above
Scenario	As the ISA onboard HMCS HALIFAX, you will be required to administer IS Equipment	
Assessment Steps	**Input:** *Part 1:* Trainees will be given a 2-scenarios that requires them to demonstrate their knowledge to administer IS Equipment. *Part 2:* Trainees are to debrief Instructors to demonstrate their knowledge of how to Conduct IS Status Brief. **Performance**: Trainees will have 150 minutes to complete this EC. **Output***: Part 1:* Trainees must achieve 7/7 SATs to pass this assessment. *Part 2:* Trainees must achieve 3/3 SATs to pass this assessment. **Feedback**: Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) EC_007.02_Administer_IS_Equipment(I) 2) EC_007.02_Administer_IS_Equipment(S)		
Note(s) to Instructor		
1) Instructor shall meet with trainee within one training day to assess training deficiencies. 2) References and resources may be used to complete this assessment.		
Assessment Limitations	Nil.	
Impact of Failure	Senior instructor and EO instructor shall meet with trainee within one training day to discuss assessment deficiencies. Re-test if necessary.	

PROTECTED B (when complete)

EC 007.02 Administer IS Equipment			
Item	SAT	UNSAT	Remarks
PART 1			
3. Scenario #1			
Security Groups created to include:			

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

EC 007.02 Administer IS Equipment			
"WIN MSE DEPT"	☐	☐	
"WIN MSE MS & ABOVE"	☐	☐	
File permissions assigned to "Common Drive" to include:			
"WIN MSE DEPT" given Read & Write access to "MSE Dept" folder	☐	☐	
"WIN MSE DEPT" given Read & Write access to "MSE Work" sub-folder	☐	☐	
"WIN MSE MS & ABOVE" the ONLY group given Read & Write access to "MSE Divisional" folder	☐	☐	
4. Scenario #2			
Print Spooler service running	☐	☐	Located on File & Print
Test Page Printed	☐	☐	
PART 2			
Brief Chain of Command to include:			
Identified root cause	☐	☐	
Described troubleshooting steps taken	☐	☐	
Plan of action for future mitigation	☐	☐	If applicable

Score: /10

Trainee must achieve 10/10 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PO 007 – ADMINISTER INFORMATION SYSTEMS
PC 007 – ADMINISTER INFORMATION SYSTEMS

PC 007 – Administer Information Systems		
Scope	**Location Conditions**	**Instructional Sequence**
PO 007	NCITT	After EC 007.02

PC 007 – Administer Information Systems		
Time	**Classification**	**Instructor**
200 minutes	Unclassified	MS Nav Comm or Above
Scenario	As an ISA onboard HMCS HALIFAX, you will be required to administer equipment as well as assist with IS briefs and reports.	
Assessment Steps	**Input** – *Part 1:* Trainees will be given 2-scenarios that requires them to demonstrate their knowledge to administer troubleshooting procedures on IS equipment. *Part 2:* Trainees will debrief the instructor on one of the scenarios. **Performance** – Trainees will have 200 minutes to complete this assessment. **Output** – *Part 1* **-** Trainees must achieve 10/10 SATs to pass this assessment. *Part 2:* Trainees must achieve 3/3 SATs to pass this assessment. **Feedback** – Trainees will receive feedback from the instructional staff on or before next training day.	
Instructor Resources		
1) PC_007_Administer_Information_Systems(I) 2) PC_007_Administer_Information_Systems(S)		
Note(s) to Instructor		
1) References and resources may be used to complete this assessment. 2) It is recommended to have another instructor to assist with the assessment. 3) It is at the discretion of the instructors as to where the trainees will wait. 4) In preparation for part 2, Instructor will draw names to determine the order of sequence. It is recommended to have another room to conduct the debriefs.		
Assessment Limitations	Due to infrastructure, a written assessment is used to assess this PO.	
Impact of Failure	IAW NPTORD 4500-54.	

PROTECTED B (when complete)

PC 007 ADMINISTER INFORMATION SYSTEMS			
Item	SAT	UNSAT	Remarks
1. Administer IS Troubleshooting procedures			
Trouble ticket created to include:			
Specified network affected	☐	☐	
Possible root cause identified	☐	☐	
Listed number of users affected	☐	☐	
Described initial investigation	☐	☐	
Described troubleshooting steps taken	☐	☐	
Included active alarms and errors	☐	☐	
Specified documentation utilized	☐	☐	If applicable
2. Administer IS Equipment			
Virtual Workstation #4 IP address deconflicted with File & Print Server IP address	☐	☐	IAW promulgated IP Scope documentation
Virtual Workstation able to access network resources to include:			
Common Drive access	☐	☐	
Test page printed	☐	☐	
3. Conduct IS Brief			
Brief Chain of Command to include:			
Identified root cause	☐	☐	
Described troubleshooting steps taken	☐	☐	
Plan of action for future mitigation	☐	☐	If applicable

Score: /13
The trainee must achieve 13/13 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

CHAPTER 3: ASSESSMENT OF TRAINEE

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

GENERAL

1. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. All ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

3. The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

4. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

5. Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

7. Continuous monitoring of trainee progress is required to provide the following:

8. Early warning of trainee difficulties;

9. Feedback on the effectiveness of training;

10. Information for the Training Review Board (TRB) (if required); and

11. Information to assist in the completion of the CF 377 Course Report.

12. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

13. PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

14. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

15. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

16. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

17. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

18. A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defence Subject Classification and Disposition System (DSCDS) Primary Number 4500.

ANNEX A – REFERENCES

This table lists all references used within this QSP.

#	Short Title	Long Title
A1	CFCD 107	Canadian Forces Controlled Document 107(L)
A2	CFCD 128	Formation and Fleet IM Manual
A3	CIB 009/12	Communications Information Bulletin
A4	NDSOD	National Defence Security Orders and Directives
A5	RCN RF Safety Directive S9	RCN RF Safety Directive S9
A6	SEMS	Safety & Environmental Management System
A7	SSOs	Ship Standing Orders
A8	ITSS-04	Information Technology Security Standards
A9	ITSS-06	Information Technology Security Standards
A10	NAVIS -WPIV	Naval Information System Work Period Instructions Version: Rev 5
A11	ITSM – IT Asset Tracking Process	IT Service Management – IT Asset Tracking Process Ver:1.2
A12	MOC	Manual of Ceremony
A13	INFOSEC 2	Information Security 2 - COMSEC Material Control Policy, Standards and Procedures
A14	EITSM	EITSM eLearning Modules
A15	BIS Approved Product	Approved Products
A16	NavIS Web	NavIS SOPs
A17	CAMSAR	Canadian Aeronautical and Maritime Search and Rescue Manual (B-GA-209-001/FP-001)
B1	AXP 3	Allied Naval Communication Exercises
B2	ATP 1 Vol II	Allied Maritime Tactical Signal and Maneuvering Book
B3	ATP 16	Replenishment At Sea
B4	INTERCO	International Code of Signals
B5	ACP 130	Allied Communications Publication 130(A), Communication Instructions, Signalling Procedures in the Visual Medium
C1	AIT Learner Manual	Advanced Instructional Techniques Course Learner Manual

ANNEX B – TASK LIST

This table provides an overview of MTL tasks WRT if/where they are trained within this QSP. No train substantiations are located in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
AT0000	**Duty Area A - Visual Communications**		
AT0010	Transmit tactical information by flag hoist	NT	
AT0020	Receive tactical information by flag hoist	NT	
AT0030	Supervise signalling by flag hoist	T	PO 003
AT0040	Transmit tactical information by flashing light	NT	
AT0050	Receive tactical information by flashing light	NT	
AT0060	Supervise signalling by flashing light	NT	
BT0000	**Duty Area B - Satellite Op Centre**		
BT0010	Configure military satellite circuits	NT	
BT0020	Configure civilian satellite circuits	NT	
BT0030	Monitor military satellite connectivity	T	PO 002
BT0040	Monitor civilian satellite connectivity	T	PO 002
BT0050	Troubleshoot military satellite connectivity	T	PO 002
BT0060	Troubleshoot civilian satellite connectivity	T	PO 002
BT0070	Assign military satellite circuits	NT	
BT0080	Assign civilian satellite circuits	NT	
BT0090	Manage military satellite assets	NT	
BT0100	Manage civilian satellite assets	NT	
BT0110	Coordinate the planning of satellite communication mission assignments	NT	
BT0120	Plan satellite communication mission assignments	NT	
BT0130	Monitor military satellite circuit assignments	NT	
BT0140	Monitor civilian satellite circuit assignments	NT	
CT0000	**Duty Area C - Operations**		
CT0010	Communicate ship's movements to VTMS	NT	
CT0020	Transmit information by radiotelephone	NT	
CT0030	Receive information by radiotelephone	NT	
CT0040	Encode tactical signals	NT	
CT0050	Decode tactical signals	NT	
CT0060	Supervise encoding/decoding of tactical signals	NT	
CT0070	Plot fleet manoeuvring signals	NT	
CT0080	Supervise plotting of fleet manoeuvring signals	NT	
CT0090	Advise command on fleet manoeuvring signals	NT	
CT0100	Coordinate TG-level fleet manoeuvring	NT	
CT0110	Plot formations	NT	
CT0120	Plot screens	NT	
CT0130	Report tactical signals	NT	

CT0140	Supervise the reporting of tactical signals	NT	
CT0150	Draft OPTASK Maritime Interdiction Operations communications procedures	NT	
CT0160	Establish comms with vessels of interest (MIO)	NT	
CT0170	Question VOI (MIO)	NT	
CT0180	Direct VOI IAW command intent (MIO)	NT	
CT0190	Record MIO communications	NT	
CT0200	Supervise MIO communications	NT	
CT0210	Manage MIO communications	NT	
DT0000	**Duty Area D - Communication Support**		
DT0010	Flash up communications equipment (ship)	NT	
DT0020	Shut down communications equipment (ship)	NT	
DT0030	Direct the flash up/shut down of communications equipment (ship)	T	PO 002
DT0040	Supervise the flash up/shut down of communications equipment (ship)	NT	
DT0050	Manage the flash up/shut down of communications equipment (ship)	NT	
DT0060	Establish voice circuit	NT	
DT0070	Establish data link circuit	NT	
DT0080	Establish radio teletype circuit	NT	
DT0090	Establish IP network	NT	
DT0100	Supervise the establishment of communications circuits	NT	
DT0110	Transmit information by voice circuit	NT	
DT0120	Receive information by voice circuit	NT	
DT0130	Transmit information by data link circuit	NT	
DT0140	Receive information by data link circuit	NT	
DT0150	Transmit information by radio teletype circuit	NT	
DT0160	Receive information by radio teletype circuit	NT	
DT0170	Transmit information by IP network	NT	
DT0180	Receive information by IP network	NT	
DT0190	Direct transmission/reception of information by communications circuits	T	PO 002
DT0200	Supervise transmission/reception information by communications circuits	NT	
DT0210	Maintain communication logs	NT	
DT0220	Conduct over the air distribution/rekeying of cryptographic material	NT	
DT0230	Configure cryptographic equipment	NT	
DT0240	Load keymat into cryptographic equipment	NT	
DT0250	Destroy cryptographic keymat	NT	
DT0260	Perform communications checks	NT	
DT0270	Direct communications checks	T	PO 002

DT0280	Supervise communications checks	NT	
DT0290	Update communication publications	NT	
DT0300	Distribute internal messages	NT	
DT0310	Direct distribution of internal messages	T	PO 002
DT0320	Manage distribution of internal messages	NT	
DT0330	Troubleshoot communication circuits	NT	
DT0340	Direct troubleshooting of communication circuits	NT	
DT0350	Manage troubleshooting of communication circuits	NT	
DT0360	Restore communication equipment after power failures	NT	
DT0370	Coordinate communication equipment restoral after power failure	T	PO 002
DT0380	Manage communication equipment restoral after power failure	NT	
DT0390	Draft communication status messages	T	PO 001
DT0400	Review communication status messages	NT	
DT0410	Advise command on RADHAZ safety	NT	
DT0420	Advise command on communications status	NT	
DT0430	Manage ship's communications	NT	
DT0440	Generate Information Exchange Requirements	NT	
DT0450	Monitor ship's communications performance	NT	
DT0460	Identify unit communication capabilities and deficiencies	NT	
DT0470	Draft communications deficiency message	NT	
DT0480	Provide input to deployment reports (mid/post)	NT	
DT0490	Conduct SAR communication procedures	NT	
DT0500	Direct SAR communication procedures	T	PO 002
DT0510	Supervise SAR communication procedures	NT	
DT0520	Advise command on SAR communications	NT	
DT0530	Create ship's comm plan	NT	
DT0540	Prepare unit frequency clearance requests	NT	
DT0550	Flash up communications equipment (shore)	NT	
DT0560	Shut down communications equipment (shore)	NT	
DT0570	Direct the flash up/shut down of communications equipment (shore)	NT	
DT0580	Supervise the flash up/shut down of communications equipment (shore)	NT	
DT0590	Manage the flash up/shut down of communications equipment (shore)	NT	
DT0600	Relay message traffic to/from aircraft	NT	
DT0610	Coordinate SAR messaging to/from aircraft	NT	
DT0620	Configure NRS/NCS circuits using RCMS	NT	
DT0630	Direct configuration of NRS/NCS circuits using RCMS	NT	

DT0640	Supervise configuration of NRS/NCS circuits using RCMS	NT	
DT0650	Troubleshoot RCMS	NT	
DT0660	Control communication distribution using MSAX	NT	
DT0670	Direct communication distribution using MSAX	NT	
DT0680	Supervise communication distribution using MSAX	NT	
DT0690	Troubleshoot MSAX	NT	
DT0700	Distribute messages in COMMCEN	NT	
DT0710	Direct message distribution in COMMCEN	NT	
DT0720	Supervise message distribution in COMMCEN	NT	
DT0670	Direct communication distribution using MSAX	NT	
DT0680	Supervise communication distribution using MSAX	NT	
DT0690	Troubleshoot MSAX	NT	
DT0700	Distribute messages in COMMCEN	NT	
DT0710	Direct message distribution in COMMCEN	NT	
DT0720	Supervise message distribution in COMMCEN	NT	
ET0000	**Duty Area E - Information Technology**		
ET0010	Flash up IS equipment	NT	
ET0020	Shut down IS equipment	NT	
ET0030	Direct flash up/shut down of IS equipment	T	PO 002
ET0040	Supervise flash up/shut down of IS equipment	NT	
ET0050	Perform IS technical comm checks	T	PO 002
ET0060	Direct IS technical comm checks	T	PO 002
ET0070	Supervise IS technical comm checks	NT	
ET0080	Manage IS technical comm checks	NT	
ET0090	Plan the conduct of pre-sail checks of IS equipment	NT	
ET0100	Conduct pre-sail checks of IS equipment	NT	
ET0110	Supervise pre-sail checks of IS equipment	NT	
ET0120	Install software patches/upgrades	NT	
ET0130	Install software on workstation/server	NT	
ET0140	Configure software on workstation/server	NT	
ET0150	Supervise installation of software on workstation/server	T	PO 007
ET0160	Manage installation of software on workstation/server	T	PO 007
ET0170	Install IS hardware on workstation/server	NT	
ET0180	Uninstall IS hardware on workstation/server	NT	
ET0190	Configure hardware on workstation/server	NT	
ET0200	Supervise installation of hardware on workstation/server	T	PO 007
ET0210	Manage installation of hardware on workstation/server	T	PO 007
ET0220	Move servers between ship and shore	NT	
ET0230	Supervise the move of servers between ship and shore	NT	
ET0240	Manage the movement of servers between ship and shore	NT	

ET0250	Maintain inventory of IT assets	NT	
ET0260	Lifecycle IT assets	NT	
ET0270	Coordinate the life-cycling of IT assets	NT	
ET0280	Create unit webpages	NT	
ET0290	Maintain unit webpages	NT	
ET0300	Create network user accounts	NT	
ET0310	Administer network user accounts	NT	
ET0320	Manage network user accounts	T	PO 005
ET0330	Respond to trouble desk calls	NT	
ET0340	Assist clients with creation of trouble tickets	NT	
ET0350	Render 1st line assistance to trouble calls	NT	
ET0360	Maintain IS fault log	NT	
ET0370	Perform 1st line reaction to IS security incident	NT	
ET0380	Log incidents and metrics of alerts	NT	
ET0390	Troubleshoot workstation faults	NT	
ET0400	Supervise troubleshooting of workstation faults	T	PO 007
ET0410	Rectify workstation faults	NT	
ET0420	Supervise rectification of workstation faults	T	PO 007
ET0430	Troubleshoot network disruptions/outages	NT	
ET0440	Supervise troubleshooting of network disruptions/outages	NT	
ET0450	Rectify network disruptions/outages	NT	
ET0460	Supervise rectification of network disruptions/outages	NT	
ET0470	Report network disruptions/outages	NT	
ET0480	Write IS disaster response plan	NT	
ET0490	Implement IS disaster response plan	NT	
ET0500	Perform IS backup	NT	
ET0510	Supervise backing up of IS	NT	
ET0520	Restore IS from backup	NT	
ET0530	Supervise restoral of IS from backup	NT	
ET0540	Monitor trouble ticket rectification status	NT	
ET0550	Monitor network connectivity	NT	
ET0560	Confirm network bandwidth meets requirements	NT	
ET0570	Prepare IS status brief	T	PO 007
ET0580	Implement River City conditions	NT	
ET0590	Supervise implementation of River City conditions	NT	
ET0600	Manage ship's Information Systems	NT	
ET0610	Advise command on Information Systems status	NT	
ET0620	Advise command on Information Systems capabilities	NT	
FT0000	**Duty Area F - Information Systems Security**		
FT0010	Conduct IS security assessments	NT	
FT0020	Conduct auditing and security compliance assessment	NT	
FT0030	Brief new users on IS Security	NT	

FT0040	Manage unit IS Security education programme	NT	
FT0050	Respond to IS security incident	T	PO 007
FT0060	Coordinate response to IS security incident	NT	
FT0070	Report initial IS security incident	T	PO 007
FT0080	Conduct investigation of IS security incident	NT	
FT0090	Supervise the investigation of IS security incident	NT	
FT0100	Log IS Security incident	NT	
FT0110	Draft Information System Security Incident report	NT	
FT0120	Submit Information System Security Incident Report	NT	
FT0130	Maintain ISSIR log and file	NT	
FT0140	Ensure IS equipment is in compliance with TCI direction	NT	
FT0150	Enforce compliance of IS equipment with TCI direction	NT	
FT0160	Contribute to emergency destruction plan	T	PO 002
FT0170	Destroy cryptographic equipment	T	PO 002
FT0180	Destroy communication equipment	T	PO 002
FT0190	Identify new IS requirements	NT	
FT0200	Submit requests for IS configuration management changes (RFC)	NT	
FT0210	Manage change management procedures	NT	
FT0220	Prepare Threat & Risk Analysis documents (TRA)	NT	
FT0230	Submit Threat & Risk Analysis documents (TRA)	NT	
FT0240	Prepare Technical Compliance Inspection requests (TCI)	NT	
FT0250	Submit requests for Technical Compliance Inspections (TCI)	NT	
FT0260	Prepare Security Assessment and Authorization documents (SA&A)	NT	
FT0270	Submit Security Assessment and Authorization documents (SA&A)	NT	
FT0280	Write unit Information System Security orders	NT	
FT0290	Plan cyber defense strategy for shipboard networks	NT	
FT0300	Implement cyber defense strategy for shipboard networks	NT	
FT0310	Advise command on defense strategy for shipboard networks	NT	
FT0320	Plan cyber awareness training	NT	
FT0330	Implement cyber awareness training	NT	
FT0340	Conduct response to cyber-attack/incident on shipboard networks	NT	
FT0350	Supervise response to cyber-attack/incident on shipboard networks	NT	

FT0360	Report on cyber response to attack/incident on shipboard networks	NT	
FT0370	Enforce unit compliance with IS security orders and directives	NT	
FT0380	Enforce compliance with River City state	T	PO 002
FT0390	Advise command on IS Security	NT	
GT0000	**Duty Area G - Communications Planning**		
GT0010	Plan communications checks	NT	
GT0020	Plan communications exercises and training	NT	
GT0030	Assess communications exercises and training	NT	
GT0040	Manage communications exercises and training	NT	
GT0050	Coordinate national reviews of communication publications	NT	
GT0060	Coordinate updating of RCN communications publications	NT	
GT0070	Participate in allied communications publications (ACP) working groups and boards	NT	
GT0080	Provide input into IS security policies/directives	NT	
GT0090	Implement OPTASK IM direction	NT	
GT0100	Manage radio licences	NT	
GT0110	Coordinate frequency clearance requests	NT	
GT0120	Obtain allied/foreign approval for frequency clearance requests	NT	
GT0130	Manage information exchange requirements (IER)	NT	
GT0140	Manage communication equipment for Submarine Directorate (NATO)	NT	
GT0150	Manage requests for Technical Compliance Inspections (TCI)	NT	
GT0160	Manage submission of Security Assessment and Authorization documents (SA&A)	NT	
GT0170	Manage submission of Threat & Risk Analysis documents (TRA)	NT	
GT0180	Advise command on new IS requests	NT	
GT0190	Advise command on naval exercise communication requirements	NT	
GT0200	Draft communication exercise messages	NT	
GT0210	Advise command on naval operational communication requirements	NT	
GT0220	Draft communication operational messages	NT	
GT0230	Request military satellite circuits	NT	
GT0240	Request civilian satellite circuits	NT	
GT0250	Coordinate DAMA/TBA requirements with USN	NT	
GT0260	Prepare communications plans	NT	

GT0270	Prepare Fleet/Formation frequency clearance requests	NT	
GT0280	Monitor communications deficiency messages	NT	
GT0290	Coordinate resolution of communication deficiencies	NT	
GT0300	Assess deployment report items (mid/post)	NT	
GT0310	Draft communications Statements of Operational Deficiencies (SOCD)	NT	
GT0320	Draft documents to address communications deficiencies	NT	
GT0330	Advise command on naval communications	NT	
HT0000	**Duty Area H - Instruction & Development**		
HT0010	Prepare students for formal progress review boards	NT	
HT0020	Manage course resources	NT	
HT0030	Manage course schedule	NT	
HT0040	Monitor course files	NT	
HT0050	Manage Training Establishment facilities	NT	
HT0060	Manage Training Establishment personnel	NT	
HT0070	Debrief students on performance	NT	
HT0080	Supervise occupational instructors	NT	
HT0090	Manage course staff and students	NT	
HT0100	Recommend changes to TP and lessons	NT	
HT0110	Develop lesson plans and associated courseware	NT	
HT0120	Supervise instructors	NT	
HT0130	Instruct occupational practical (skill) development lessons	NT	
HT0140	Instruct occupational theory lessons	NT	
HT0150	Conduct instructor feedback sessions	NT	
HT0160	Develop student assessments	NT	
HT0170	Instruct occupational theory (knowledge) lessons	NT	
HT0180	Instruct occupational practical skill development lessons	NT	
HT0190	Instruct unit occupational theory/Practical (knowledge/skill) lessons	NT	
HT0200	Advise instructional staff on training technologies	NT	
HT0210	Plan Training Staff Professional Development	NT	
HT0220	Conduct Training Staff Professional Development	NT	
HT0230	Participation in Progress Review and TRBs	NT	
HT0240	Maintain the accuracy of the QSP	NT	
HT0250	Coach Instructors	NT	
HT0260	Administer course critique process	NT	
HT0270	Manage trainee assessment	NT	
HT0280	Advise on development of courseware (lesson plans etc.)	NT	
HT0290	Track action items identified during course evaluation	NT	

HT0300	Advise on training contracts and proposals	NT	
HT0310	Conduct Instructional Analysis (method, media, environment)	NT	
HT0320	Convene, coordinate and support to QSP Writing Board	NT	
HT0330	Conduct instruction monitoring for QSP compliance	NT	
HT0340	Ensure instructors are monitored	NT	
HT0350	Develop Assessment tools (PC and EC)	NT	
HT0360	Recommend improvements to training	NT	
HT0370	Plan QSP Writing Boards	NT	
HT0380	Establish a prioritised schedule for QSP writing boards	NT	
HT0390	Audit course reports	NT	
HT0400	Analyse success and failure data	NT	
HT0410	Conduct Programme evaluation (alignment of docs, availability of training materials	NT	
HT0420	Coordinate Training opportunities with Trainer providers	NT	
HT0430	Evaluate PLAR applications	NT	
HT0440	Support TNA Process	NT	
HT0450	Orient new Staff with IT&E-CT-School policy and procedures	NT	
HT0460	Support the development of training contract (RFP`s and sows)	NT	
IT0000	**Duty Area I - Training**		
IT0010	Manage on-the-job-training (OJT) programs at the unit level.	NT	
IT0020	Instruct unit occupational practical skill development lessons	NT	
IT0030	Instruct unit occupational theory lessons	NT	
IT0040	Develop (amend) unit training lesson plan and associated courseware	NT	
IT0050	Manage Training Plans	NT	
IT0060	Maintain Operational Readiness of Assigned Section	NT	
JT0000	**Duty Area J - Career Management**		
JT0010	Prepare occupational posting plots	NT	
JT0020	Brief personnel on occupational career management issues	NT	
JT0030	Manage occupation qualification standards	NT	
JT0040	Determine occupational cost move requirements	NT	
JT0050	Carry out occupational personnel interviews	NT	
JT0060	Prepare occupational personnel files for selection boards	NT	
JT0070	Prepare occupational personnel files administrative reviews	NT	

JT0080	Select occupational personnel for training	NT	
JT0090	Advise on occupational personnel issues	NT	
JT0100	Prepare occupational personnel files for release from CAF	NT	
JT0110	Select occupational personnel for postings	NT	
JT0120	Brief annual military occupation review (AMOR) boards	NT	
JT0130	Prepare occupational personnel messages	NT	
KT0000	**Duty Area K - Standards**		
KT0010	Manage course reports	NT	
KT0020	Monitor course conduct compliance with TP	NT	
KT0030	Monitor assessment of students	NT	
KT0040	Advise Chain of Command (CoC) on validation results	NT	
KT0050	Determine test requirements	NT	
KT0060	Monitor compliance of lesson plans and associated courseware with training plan	NT	
KT0070	Supervise assessment of student performance	NT	
KT0080	Develop test items	NT	
KT0090	Approve amended lesson plans and associated courseware	NT	
KT0100	Assess student performance	NT	
KT0110	Conduct course evaluations	NT	
KT0120	Advise on training standards	NT	
KT0130	Conduct Validations	NT	
KT0140	Approve amended qualification standards	NT	
KT0150	Advise the Chain of Command (CoC) on evaluation results	NT	
KT0160	Monitor compliance of qualification standards with occupational specifications	NT	
KT0170	Monitor instructor performance	NT	
KT0180	Manage occupational instruction	NT	
KT0190	Conduct occupational training standards activities	NT	
KT0200	Supervise occupational training standards personnel	NT	
KT0210	Manage Training Plans	NT	
LT0000	**Duty Area L - Submariner Tasks**		
LT0010	Update damage control plot	NT	
LT0020	Communicate damage control status	NT	
LT0030	Monitor bulkhead 56	NT	
LT0040	Monitor interior atmospheric conditions	NT	
Added Tasks			
	Draft communications messages	T	PO 001
	Plan communications exercises and training	T	PO 001
	Conduct communications exercises and training	T	PO 001

	Report communications status	T	PO 001
	Direct CCR operations	T	PO 002
	Direct visual communications	T	PO 003
	Conduct duties of flag deck supervisor	T	PO 003
	Administer information systems	T	PO 004
	Administer IS equipment	T	PO 004
	Assist with IS briefs and reports	T	PO 004
	Administer server virtualization (VMware)	T	PO 005
	Create templates and servers using VMware (vCenter)	T	PO 005
	Deploy templates and servers using VMware	T	PO 005
	Install vCenter	T	PO 005
	Configure vCenter	T	PO 005
	Configure vSwitches and VMware roles and permissions	T	PO 005
	Create domain controllers and join the domain	T	PO 005
	Utilize datastores	T	PO 005
	Add the DNS server role	T	PO 005
	Administer server management	T	PO 006
	Manage the print service	T	PO 006
	Create scripts and profiles	T	PO 006
	Manage the Exchange Server	T	PO 006
	Manage Network Monitoring Tools	T	PO 007
	Supervise Liaison with Local IS Authorities	T	PO 007
	Providing Technical Overview of IS Equipment	T	PO 007

ANNEX C – RECORD OF DECISIONS

The following summary provides insight to the decisions and reasoning resulting in this QSP.

Dates: 18 Jan 2021 - 27 Sep 2021

Location: CFB Halifax Stadacona Building S-15 Rm 323

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Meyers	R	Lt	00211	NTDC(A)
Scheller	L	CPO2	00299	NTDC(A)
Theriault	M	PO1	00299	NTDC(A)
Bunleon	K	PO2	00299	NTDC(A)
Aikens	G	MS	00299	NTDC(A)
Schmidt	D	S1	00299	NTDC(A)

Working Group Objective:

1. This group was initialized as the result of a Tiger Team meeting between coasts that took place in Jan 2020.

2. Legacy and consistent stop gap training program modifications have caused revisionist QSPs which incorporated legacy MTL tasks, updated MTL tasks, draft MTL tasks, and unique tasks that have never been approved for addition. As such, CNPTG authorized Nav Comm Draft MTL V.5 as the baseline for all Nav Comm QSPs until such time that tasks are reorganized following a Combat Operator OA.

3. This document is the second in a series of revisions of Nav Comm QSPs that aims to improve, modernize, and standardize training for the trade.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
AT0030	Supervise Signalling by flag hoists	Trained by virtue of On the job experience
AT0060	Supervise Signalling by Flashing Light	USQ
BT0010	Configure Military Satellite circuits	Taught in previous QL course
CT0020	Transmit information by radiotelephone	Taught in previous QL course
CT0030	Receive information by radiotelephone	Taught in previous QL course
CT0040	Encode tactical signals	Taught in previous QL course
CT0050	Decode tactical signals	Taught in previous QL course
CT0060	Supervise encoding/decoding of tactical signals	Will be taught at PO2 level
CT0070	Plot fleet manoeuvering signals	Taught in previous QL level
CT0080	Supervise the plotting of fleet manoeuvering signals	Will be taught at PO2 level
CT0090	Advise command on fleet manoeuvering signals	Will be taught at PO2 level
CT0110	Plot Formations	Taught in previous QL course
CT0120	Plot Screens	Taught in Previous QL level
CT0130	Report tactical signals	Taught in previous QL level
CT0140	Supervise the reporting of tactical signals	Will be taught at PO2 level
CT0200	Supervise MIO communications	Will be taught at PO2 level
DT0040	Supervise the flash up/shut down of communications equipment	Will be taught at PO2 level
DT0060	Establish voice circuit	Taught in previous QL course
DT0070	Establish data link circuit	Taught in previous QL course
DT0080	Establish radio teletype circuit	Taught in previous QL course
DT0090	Establish IP network	Taught in previous QL course

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

DT0100	Supervise the establishment of communications circuits	Will be taught at PO2 level
DT0170	Transmit information by IP network	Taught in previous QL course
DT0180	Receive information by IP network	Taught in previous QL course
DT0200	Supervise transmission/reception information by communications circuits	Will be taught at PO2 level
DT0210	Maintain communication logs	Taught in previous QL level
DT0220	Conduct over the air distribution/rekeying of cryptographic	Taught in previous QL level
DT0230	Configure cryptographic equipment	Taught in previous QL course
DT0240	Load keymat into cryptographic equipment	Taught in previous QL course
DT0250	Destroy cryptographic keymat	Taught in previous QL course
DT0260	Perform communications checks	Taught in previous QL course
DT0280	Supervise communication checks	Will be taught at PO2 level
DT0290	Update communication publications	Taught in previous QL course
DT0310	Direct distribution of internal messages	Will be taught at PO2 level
DT0310	Direct distribution of internal messages	Will be taught at PO2 level
DT0330	Troubleshoot communication circuits	Taught in previous QL course
DT0340	Direct troubleshooting of communication circuits	Will be taught at PO2 level
DT0360	Restore communication equipment after power failures	Will be taught at PO2 level
DT0370	Coordinate communication equipment restoral after power failure	Will be taught at PO2 level
DT0410	Advise command on RADHAZ safety	Will be taught at PO1 level

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

DT0420	Advise command on communications status	Will be taught at PO1 level
DT0450	Monitor ship's communications performance	Will be taught at PO2 level
DT0460	Identify unit communication capabilities and deficiencies	Will be taught at PO2 level
DT0490	Conduct SAR communication procedures	Taught in previous QL course
DT0550	Flash up communication equipment(shore)	Unit specific (NRS)
DT0560	Shut down communications equipment (shore)	Unit specific (NRS)
DT0570	Direct the flash up/shut down of communication equipment (shore)	Unit specific (NRS)
DT0600	Relay message traffic to/from aircraft	AGA(NRS) specific duty
DT0610	Coordinate SAR messaging to/from aircraft	AGA(NRS) specific duty
ET0010	Flash up IS equipment	Taught in previous QL course
ET0020	Shutdown IS equipment	Taught in previous QL course
ET0030	Direct flash up/shut down of IS equipment	Experience based training
ET0040	Supervise flash up/shutdown of IS equipment	Will be taught at PO2 level
ET0050	Perform IS technical comm checks	Taught in previous QL course
ET0060	Direct IS technical comm checks	Experience based training
ET0070	Supervise IS technical comm checks	Will be taught at PO2 level
ET0080	Manage IS technical comm checks	Will be taught at PO1 level
ET0090	Plan the conduct of pre-sail checks of IS equipment	Taught in previous QL course

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ET0100	Conduct pre-sail checks of IS equipment	Taught in previous QL course
ET0110	Supervise pre-sail checks of IS equipment	To be taught at PO2 level
ET0120	Install software patches/upgrades	Taught in previous QL course
ET0130	Install software on workstation/server	Taught in previous QL course
ET0140	Configure software on workstation/server	Taught on previous QL course
ET0160	Manage installation of software on workstation/server	Will be taught at PO2 level
ET0170	Install IS hardware on workstation/server	Taught in previous QL course
ET0180	Uninstall IS hardware on workstation/server	Taught in previous QL course
ET0190	Configure hardware on workstation/server	Taught in previous QL course
ET0210	Manage installation of hardware on workstation/server	To be taught at PO2 level
ET0220	Move servers between ship and shore	To be taught at PO2 level
ET0230	Supervise the move of servers between ship and shore	To be taught at PO2 level
ET0240	Manage the movement of servers between ship and shore	To be taught at PO1 level
ET0250	Maintain inventory of IT assets	Taught in previous QL course
ET0260	Lifecycle IT Assets	Taught in previous QL course
ET0270	Coordinate the life-cycling of IT assets	Taught in previous QL course
ET0280	Create unit webpages	Not a Nav Comm task, recommend removal from MTL
ET0290	Maintain unit web pages	Not a Nav Comm task, recommend removal from MTL

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ET0300	Create network user accounts	Taught in previous QL course
ET0310	Administer network user accounts	Taught in previous QL course
ET0330	Respond to trouble desk calls	Unit specific (NOC)
ET0340	Assist clients with creation of trouble tickets	Unit Specific (NOC)
ET0350	Render 1st line assistance to trouble calls	Taught in previous QL course
ET0360	Maintain IS fault log	To be taught at PO1 level
ET0370	Perform 1st line reaction to IS security incident	Taught in previous QL course
ET0380	Log incidents and metrics of alerts	Taught in Kingston phase
ET0390	Troubleshoot workstation faults	Taught in previous QL course
ET0400	Supervise troubleshooting of workstation faults	Experience based training
ET0410	Rectify workstation faults	Taught in previous QL course
ET0420	Supervise rectification of workstation faults	Experience based training
ET0430	Troubleshoot network disruptions/outages	Taught in previous QL course
ET0440	Supervise troubleshooting of network disruptions/outages	Experience based training
ET0450	Rectify network disruptions/outages	Taught in previous QL course
ET0460	Supervise rectification of network disruptions/outages	Experience based training
ET0480	Write IS disaster response plan	
ET0490	Implement IS disaster response plan	To be taught at PO1 course
ET0500	Perform IS backup	Taught in previous QL course
ET0510	Supervise backing up of IS	Experience based training
ET0520	Restore IS from backup	Taught in previous QL course
ET0530	Supervise restoral of IS from backup	Experience based training

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ET0540	Monitor trouble ticket rectification status	Unit Specific (NOC)
ET0550	Monitor Network Connectivity	Taught in previous QL course
ET0560	Confirm network bandwidth meets requirements	Covered in ET0550 (remove from MTL?)
ET0580	Implement River City conditions	Taught on previous QL course
ET0590	Supervise implementation of River City conditions	To be taught at PO1 level
ET0600	Manage ship‘s information systems	To be taught at PO1 level
ET0610	Advise command on IS status	To be taught at PO1 level
ET0620	Advise command on IS capabilities	To be taught at PO1 level
FT0030	Brief new users on IS Security	To be taught at PO1 level
FT0080	Conduct investigation of IS security incident	To be taught at PO1 level
FT0100	Log IS security incident	To be taught at PO1 level
FT0110	Draft Information System Security Incident report	To be taught at PO1 level
FT0130	Maintain ISSIR log and file	To be taught at PO1 level
FT0140	Ensure IS equipment is in compliance with TCI direction	Taught in previous QL course
FT0150	Enforce compliance of IS equipment with TCI direction	Taught in previous QL course
FT0160	Contribute to emergency destruction plan	Taught in previous QL course
FT0250	Submit requests for Technical Compliance Inspections (TCI)	Taught in previous QL course
FT0300	Implement cyber defense strategy for shipboard networks	Not a Nav Comm task, recommend removal from MTL
FT0330	Implement cyber awareness training	Not a Nav Comm task, recommend removal from MTL

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

FT0340	Conduct response to cyber-attack/incident on shipboard networks	Not a Nav Comm task, recommend removal from MTL
FT0360	Report on cyber response to attack/incident on shipboard networks	Not a Nav Comm task, recommend removal from MTL
FT0380	Enforce compliance with River City State	Taught in previous QL course
FT0390	Advise command on IS Security	To be taught at PO1 level
GT0010	Plan Communications Checks	Taught throughout various POs in various QL levels
GT0020	Plan Communications Exercises and Training	To be taught at PO2 level
GT0030	Assess Communications Exercises and Training	To be taught at PO2 level

Discussions / Decisions:

4. Upon convening of this Working Group (WG), Subject Matter Experts (SMEs) along with PO2 Wiseman of the Comms School came together to review Train/No Train items. This was done to better identify the knowledge needed and appropriate for this Qualification Level (QL).

5. Due to limitations and deficiencies caused by the current QL5B QSP, instructors needed to build their own training content for the course. From these circumstances this course was widely agreed to be in potentially the worst shape of the legacy Nav Comm trade courses. A greater part of the content and instruction requires re-organization with other qualification levels. Other tasks laid out within are simply experience based training.

6. Task analysis resulted initially in 4 performance objectives as well as a separate IT phase in Kingston.

Commented [a1]: Needs to have amplifying information

7. WG consulted SMEs in the IT field for consultation WRT bringing the Kingston phase training in-house at the NCITT. With support from PO2 Quesnelle, the WG conducted task analysis of the Kingston phase IT program, pulling the content of any Nav Comm QL5B related tasks. This resulted in the creation and implementation of 3 additional performance objectives, bringing the total to 7.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

8. The WG, with support from the NCITT maintainers, trialed and tested all material used from the Kingston phase, ensuring compatibility with NCITT.

9. PO 003, 004 and 005 do not have tasks from MTL assigned. Tasks indicated by Training Manager as essential for job performance. Official task numbers not assigned due to OA.

10. EO 004.02 and EO 004.03 were not assigned ECs to avoid redundancy. ECs would be repetitions of the labs and would hold no training value. They are practiced several times throughout the lessons.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion

Course History:

Dates: 22 September – 9 October 2015, 0800 - 1600

Location: CFB Esquimalt CFFSE CTC RM 207/208 Board Members:

Surname	Initials	Rank	MOSID	Unit
Pinard	S	LT(N)	00207	HMCS REGINA
Sprang	L	LT(N)	00211	CFFSE
Froment	D	CPO2	00299	NPTG HQ Halifax

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

Macintosh	C	PO1	00299	CFP
Francis	A	PO2	00299	CFFSE
Lacroix	M	PO2	00299	CFNOS
Nadeau	D	PO2	00299	CFFSE
Franklin	J	MS	00299	CFNOS
Merrifield	A	MS	00299	CFFSE

Board Objective:

The objective of this QSP review board is to restructure the Nav Comm QL5B to align its relevance with recent changes to the Job-Based Specifications (JBS) and incorporate changes due to out-service training/contracts demands.

Day 1 – 22 Sep 15
Board members reviewed Scalar for QL5B. The TDO reviewed past course critiques, which included comments:

- TACLANE was applicable, as well as KIV 7M.
- Improve/change ShipLAN. They liked the simulation and asked for more hands- on.
- Add more about DRIMS, RDIMS, LRA (PKI), Monitor Mass, Endpoint Protection, Entrust dialogue box, DVPNI.
- They would like course shortened,
- Increased hands-on with server/ShipLAN simulation, and
- ITSM.

The board members described what they expect a Nav Comm, acting in a supervisory role, to know upon completion of the QL5B course:

- COMSEC,
- Direct Equipment Flash up,
- SHOW Bridge/CCR/OPS Comms,

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

- ISA (Create/Delete accounts, back up, image computers, basic troubleshooting, draft Trouble Tickets under supervision)
- Bandwidth and OPSEC Management
- Draft messages

The Board members identified which of the Nav Comm JBS Tasks were to be trained/not trained.

Board members organized the various JBS Tasks into functional groups, and sequencing. Renaming the POs as required.

There was discussion about what DT0060/65 and AT00218 meant. The CFFSE
Communications Training Chief clarified that tasks DT0060/65 were about developing training plans for their section and for multiple unit exercises.

AT00218 was discussed regarding how to train "manage EDP" on Major Warships vs Minor War Vessels (MWV), on MWV you are in charge of the entire ship, but on Major Warships it is just for your section. The CFFSE Communications Training Chief also discussed AT00218, so that it includes MWV considerations.

The Board members created all the POs. Annexes A and B were initiated.

Time	Discussed Topic
Day 1 – 22 Sep 15	
0800	SCALAR, review what 5B trainees need to know, past course critiques. Made train/no train list
1000	Began separating into POs
1300	Continued working on POs, Annexes A and B initiated

Specification Task – No Train Decisions:

Task No	Occupational Requirements	Explanation
		Previously trained and performed often
AT0081	Communicate information by Flag Hoist	Previously trained and performed often
AT0082	Communicate information by Flashing Light	Previously trained and performed often

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

AT0083	Communicate information by Network Data means	Previously trained and performed often
AT0084	Communicate information by Radio Telephone	Previously trained and performed often
AT0085	Communicate information by Inter-ship	Previously trained and performed often
AT0086	Communicate information by Ship-Shore	Previously trained and performed often
AT0087	Communicate information by Shore-Ship	Previously trained and performed often
AT0088	Communicate information by CRATT	Previously trained and performed often
AT0089	Communicate information by Broadcast	Previously trained and performed often
AT0091	Communicate with commercial shipping and VTMS	Previously trained and performed often
AT0155	Prepare Communication Plans (COMPLANs)	Previously Trained
AT0160	Prepare Frequency Plans	Previously Trained
AT0165	Extract information from COMPLANS and Frequency Plans	Previously trained and performed often
AT0172	Apply Frequency Management Principles	Previously Trained

AT0180	Select communication frequencies	Previously trained and performed often
AT0190	Prepare frequency clearance requests	Previously Trained
AT0205	Draft Messages for Transmission	Previously trained and performed often
AT0206	Employ Pyrotechnic devices	Previously trained and performed often
AT0207	Direct the processing of communications	Previously trained and performed often
AT0209	Brief Flag Ceremonial Procedures	Previously trained and performed often
AT0210	Coordinate Flag Ceremonial Procedures	Previously trained and performed often
AT0211	Update communications publications	Previously trained and performed often
AT0212	Muster CSDs and COMSEC Material	Previously trained and performed often
AT0213	Implement OPTASK IM direction	Previously Trained
AT0215	Draft Operational Messages	Previously trained and performed often
AT0216	Advise Command on bandwidth management	Wrong level. Revised task name to ‘Advise the Chain of Command on bandwidth management’ (AT0221)

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

AT0217	Advise Command on shipboard IS capabilities	Wrong level. Revised task name to 'Advise the Chain of Command on shipboard IS capabilities' (AT0220)
AT0250	Maintain communication logs	Previously trained and performed often
AT0260	Coordinate equipment restoration during power failures	Previously trained and performed often
AT0275	Operate point-to-point circuits (i.e. PC to PC comms)	Previously Trained and Obsolete
BT0050	Describe RF Safety procedures	Previously trained and performed often
BT0051	Direct RF Safety procedures	Previously trained and performed often
BT0054	Initialize/Shut down UHF (LOS) Communications Equipment	Previously trained and performed often
BT0055	Initialize/Shut down VHF Communications Equipment	Previously trained and performed often
BT0056	Initialize/Shut down HF Communications Equipment	Previously trained and performed often
BT0057	Initialize/Shut down Satellite Communications Equipment	Previously trained and performed often
BT0058	Initialize/Shut down Communications Control Monitoring System (CCMS) Equipment	Previously trained and performed often
BT0059	Initialize/Shut down Cryptographic equipment	Previously trained and performed often
BT0060	Initialize/Shut down Message Processing system equipment	Previously trained and performed often
BT0062	Direct the Initialization/Shut down of Communications Equipment	Previously trained and performed often
BT0065	Troubleshoot UHF (LOS) Communications Equipment Faults	Previously trained and performed often
BT0066	Troubleshoot VHF Communications Equipment Faults	Previously trained and performed often

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

BT0067	Troubleshoot HF Communications Equipment Faults	Previously trained and performed often
BT0068	Troubleshoot Satellite Communications Equipment Faults	Previously trained and performed often
BT0069	Troubleshoot Communications Control	Previously trained and performed often
BT0070	Troubleshoot Message Processing system equipment Faults	Previously trained and performed often
BT0071	Troubleshoot Crypto equipment faults	Previously trained and performed often

BT0072	Direct the Troubleshooting of Communications Equipment Faults	Previously trained and performed often
BT0075	Configure Ship-Shore circuits	Previously trained and performed often
BT0076	Configure Inter-ship circuits	Previously trained and performed often
BT0077	Configure Broadcast circuits	Previously trained and performed often
BT0078	Configure Military Satellite circuits	Previously trained and performed often
BT0079	Configure Commercial Satellite circuits	Previously trained and performed often
BT0080	Operate National Secure Networks	Previously trained and performed often
BT0081	Configure Circuits Within Message Processing system	Previously trained and performed often
BT0082	"Configure Communications Control Monitoring System (CCMS) Equipment"	Previously trained and performed often
BT0083	Configure Cryptographic equipment	Previously trained and performed often
BT0084	Direct the Configuration of Communications Equipment	Previously trained and performed often
BT0085	Operate Allied Secure Networks	Previously trained and performed often
BT0087	Perform Communications Checks	Previously trained and performed often
BT0088	Direct Communications Checks	Previously trained and performed often
BT0090	Operate NATO Secure Networks	Previously trained and performed often
BT0093	Load Cryptographic Equipment	Previously trained and performed often

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

BT0094	Verify Cryptographic Keymat	Previously trained and performed often
BT0095	Implement EMCON states	Previously trained and performed often
BT0098	Configure voice circuits	Previously trained and performed often
CT0001	Destroy Cryptographic Keymat	Previously trained and performed often
CT0010	Employ low grade codes	Previously trained and performed often
CT0020	Conduct Over the Air Distribution (OTAD) procedures	Previously trained and performed often
DT0015	Perform anti-jam procedures	Previously Trained
DT0020	Report Tactical signals	Previously trained and performed often
DT0021	Encode/Decode Tactical signals	Previously trained and performed often
DT0022	Advise Command on Tactical signals	Previously trained and performed often
DT0023	Plot Fleet Manoeuvring Signals	Previously trained and performed often
DT0030	Conduct Maritime Interdiction Operations communications	Previously trained and performed often
DT0031	Direct Maritime Interdiction Operations communications	Previously trained and performed often
DT0034	Interpret OPTASK Maritime Interdiction Operations Communications Procedures	Previously trained and performed often
DT0050	Conduct SAR Procedures	Previously trained and performed often
DT0051	Direct SAR Procedures	Previously trained and performed often
DT0085	Rig/strike dressing lines	Previously trained and performed often
DT0115	Plot formations, screens and dispositions	Previously trained and performed often
ET0010	Enforce Communications Security	Previously trained and performed often
ET0011	Describe COMSEC	Previously trained and performed often
ET0012	Identify COMSEC Incidents	Previously trained and performed often
ET0013	React to COMSEC Incidents	Previously trained and performed often
ET0014	Describe IS SEC	Previously trained and performed often
ET0025	Identify IS SEC incidents	Previously trained and performed often
ET0026	React to IS SEC incident	Previously trained and performed often
ET0033	Enforce IS security	Previously trained and performed often

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ET0043	Enforce physical security (Non IS)	Previously trained and performed often
FT0010	Install/Uninstall IS components	Previously trained and performed often
FT0012	Monitor IS hardware installation and upgrades	Previously trained and performed often
FT0016	Maintain IS components	Previously trained and performed often
FT0022	Maintain network connectivity	Previously trained and performed often
FT0037	Report network disruptions/outages	Previously trained and performed often
FT0040	Install and configure OS on a computer	Previously trained and performed often
FT0041	Install and configure Standalone applications	Previously trained and performed often
FT0042	Install and configure Network applications	Previously trained and performed often
FT0043	Apply software upgrades and patches	Previously trained and performed often
FT0044	Install and configure local IS peripherals	Previously trained and performed often
FT0045	Install and configure network IS peripherals	Previously trained and performed often
FT0046	Troubleshoot workstation faults	Previously trained and performed often
FT0047	Rectify workstation faults	Previously trained and performed often
FT0048	Initialize/Shut down Network equipment	Previously trained and performed often
FT0049	Document IS Faults	Previously trained and performed often
FT0060	Administer User accounts	Previously trained and performed often
FT0063	Operate Remote Servers	Previously trained and performed often
FT0064	Employ IT/IS Change Management	Previously trained and performed often
FT0070	Conduct network Pre-sail checks	Previously Trained
FT0080	Establish preventative IS maintenance schedule	Previously Trained
FT0081	Perform network health checks	Previously trained and performed often
FT0082	Control IS equipment inventory and distribution	Previously trained and performed often
FT0083	Lifecycle IS components	Previously trained and performed often
FT0084	Maintain integrity, confidentiality, availability of IS Networks	Previously trained and performed often
FT0086	Troubleshoot LAN faults	Previously trained and performed often
FT0087	Rectify LAN Faults	Previously Trained
FT0088	Troubleshoot external network connectivity	Previously Trained

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

FT0089	Rectify external network connectivity	Previously Trained

Day 2 - 23 Sep 2015
The Board members separated into groups and went over the POs that weren't created during day 1. Annex A was reviewed and vetted of references that weren't applicable. The CFFSE Communications Training Chief came in to be briefed on the newly created POs.

Time	Discussed Topic
Day 2 – 23 Sep 15	
0800	Review/Break down POs
1000	Vetted Annex A of irrelevant Refs
1040	CPO2 Cherneff came in to be briefed on how the POs were set up
1300	Reviewed Annex A, made additions/deletions
1330	Drafted POs in Chapter 2 QSP

Day 3 – 24 Sep 15
The board members revised Chapters 1 and 2 of the QSP to reflect recent changes in policies and cessation of out-sourced training. NPTG's representative believed that the board members captured the essentials in these two chapters.

The board members discussed the option to employ civilian instructors to train the basics of networking so as to provide consistency in the level and quality of networking fundamentals. After further discussions, it was decided that IT should be taught by Nav Comms, as this would provide more control of the content to that trade and would allow them to dynamically adapt to evolving IT.

Day 4 – 25 Sep 15
The board members divided themselves into three groups and began designing the EOs for POs 001, 002, 003 and 008.

Day 5 – 28 Sep 15
Reviewed day 3 and 4's ROP as a group.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

PO 001 was completed, and delivery was estimated as 8 hours via DLN. PO 003's total delivery time was expected to be 40 x 50 minutes periods. Work on PO 002, 003 and 008 is ongoing.

Day 6 – 29 Sep 15
Board members completed EOs for POs 002, 003, 004, 005 and 006.

PO 008 (Managing Shipborne Servers) was divided into 7 separate and comprehensive EOs. Some of these EOs will be assessed through Critical ECs. Board members discussed the feasibility of subdividing EOs into smaller groupings, assigning these groups to separate POs so as to reduce the volume of information within one PO. After further discussion, it was decided that since the knowledge and skill sets required by each EO was inter-related, it was best to keep all 7 EOs under one PO. The EOs are to be assessed in such a way that Critical ECs and the PC are divided into manageable sections for assessment purposes and scheduled to prevent learning fade between instruction and testing.

Day 7 – 30 Sep 15
Board members completed EOs for POs 007, 008 and 010. The board completed Annex K for
POs 001-007 and Annex D (Scalar), updated Annex A (References) and started work on Annex G (Suggested Sequencing and Timings). The board discussed how best to ensure the continuity between the Nav Comm QL 5B and 6A. One board member suggested placing the PO scalars for 5B on chart paper in a break-out room, available for reference purposes for the 6A board, which is scheduled one week after the conclusion of the 5B.

Day 8 – 1 Oct 15
The board reviewed the minutes. Board members relocated PO scalars onto chart paper for portability. Completed PO 009. Total training days were calculated for all POs. The board reviewed Annex K, where it was determined that ECs for PO 007 needed to be revisited. The board attended the NPTG town hall for the afternoon, limiting work on the QSP.

Day 9 – 2 Oct 15

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

The board expressed their concerns regarding the gap in knowledge and skills between the 'old' version of the QL5A and the new version of the QL5B. Those trainees having completed the previous version of the QL5A, who have not had sufficient ISA related experience on the job, are likely to find the new QL5B training quite challenging, as it is ISA-centric. Recommendations to having Module 4 of new QL5A (23 training days), act as 'delta training' for those lean on ISA experience were not well-received by the TE. The TE thought that trainees would have had enough experience between the 'old' QL5A and QL5B to compensate for the potential 'delta', and consequently, there was no need to provide formalized training. The board believes that due to HCM/FELEX and the current state of the Fleets (East and West), that this may not be an accurate assessment, and that it is likely that trainees will struggle to achieve the QL5B standard.

Day 10 – 5 Oct 15
During the forenoon, the board members reviewed chapter 4 as a group so as to identify and rectify issues with wording and ensuring that EOs and TPs were adequately described. While doing so it was identified that certain EOs were not described in detail. During the afternoon, the board was separated in 2-member groups and ensured that the appropriate level of details was captured for every EO so that we could assess chapter 4 as a group.

Day 11 – 6 Oct 15
The board reviewed chapter 4, POs 001 through 007 and 009.

Day 12 – 7 Oct 15
The board reviewed chapter 4, POs 008 and 010, as well as Annexes A – References and F – Acronyms and Abbreviations.

Day 13 – 8 Oct 15
The board finalized an ECL for PO 008 and a PCL for PO 004. Additionally, the board realigned Annex K's content to chapter 4 that was reviewed on 7 Oct 15.

The board compared chapter 4 of the QL5A and 5B QSPs to ensure continuity with minimal overlap between the two courses.

Additionally, the Instructional Scalars developed for the QL5B were moved in room 216. This was done so that once the QL6A QSP review board starts on 19 Oct, the newly formed board can easily gain situational awareness of where the QL5B left off and ensure that they can bridge the gap between the newly developed QL5B and the revised QL6A.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

The Instructional Scalars specific for IT for the QL5A were moved to room 206. Since tremendous changes occurred between the old and the revised version of the QL5A, the board members believed it was important to retain these scalars so as to give new board members a holistic view of the revision process.

Day 14 – 9 Oct 15
The Board reviewed Annex C – Record of Proceedings, for both the QL5A and 5B, so as to ensure accuracy of tasks completed and comprehensiveness of the discussions that lead to recommendations and decisions throughout both boards.

During the Instructional Design the member created JBS tasks as the provided listing either did not capture a specific task or did not appropriately describe a task to be completed. Therefore, we recommend the following tasks **additions** to the JBS Task List (QL5B): AT0001 - Perform the Duties of a Communications Supervisor;

- AT0220 - Advise the Chain of Command on shipboard IS capabilities;
- AT0221 - Advise the Chain of Command on bandwidth management;
- AT0222 - Explain the Emergency Destruction Plan;
- AT0223 - Coordinate the Phases of Emergency Destruction;
- AT0224 - Revise Emergency Destruction Plan;
- AT0225 - Advise the Chain of Command on Emergency Destruction Plan;
- AT0226 - Advise the Chain of Command on Communications Status;
- DT0111 - Develop a Training Plan;
- DT0112 - Coordinate Communication Exercises and Training;
- ET0051 - Manage OPSEC requirements;
- ET0052 - Describe IS Security Education requirements;
- FT0091 - Manage Network Connectivity;
- FT0092 - Maintain HAS/VDI/TTC Servers;
- FT0093 - Administer Virtual Servers;
- FT0094 - Administer Exchange Servers;
- FT0095 - Explain System Center Configuration Manager (SCCM) Server capability;
- FT0096 - Administer DRMIS Servers;

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

- FT0097 - Administer Domain Controllers;
- FT0098 - Administer Backup Servers;
- FT0099 - Administer MTWAN Networks;
- FT0100 - Manage Classified Networks;
- FT0101 - Maintain Minor War Vessels (MWV) Network Infrastructure;
- FT0102 - Manage Minor War Vessels (MWV) Network Infrastructure; FT0103 - Explain the Technical Configuration of CSNI; and FT0104 - Administer File Servers.

Lessons Learned

Board Composition

1. Retaining the same personnel between the QL 5A and 5B was very effective and provided continuity especially when aligning the course content. Repetition was also kept to a minimum.
2. Representation from all aspects of the occupation should be included when staffing QSP boards. We were fortunate to have a diverse group that included a PRes member, Standards, IT expertise and CCR/Bridge expertise. This enabled the board to have a wide breadth of knowledge with which to create the QSP. As communications are becoming more reliant on technology, so does the training of the Nav Comm occupation. Therefore, it is critical that QSP review/writing boards have SMEs in IT and its direct application within the RCN.
3. Expertise amongst NPTG representatives was lean. Higher level expectations were not conveyed to the board, and at times, they acted as SMEs rather than as NPTG representatives. A non-trade, honest broker might have proven more effective.
4. The chairperson should not be of the same MOSID as those on the board, nor should they have personal interest in the direction the board takes.

Documentation

1. Tasks within the JBS Task list for 5A and 5B were quite vague and omitted important tasks. The board was forced to create a number of tasks to capture the specific aspects of the tasks for those levels. The output of the JBS reflected a lack of clarity as to what the more junior jobs actually are. JBS board composition should include members that have recent experiences performing the tasks at that level.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

2. QSP documentation formatting provided by NPTG was difficult to use and made inputting information very awkward. Repetition within each EO also seemed to be redundant. Streamlining the document would be of benefit for the board members and the instructional staff.
3. Appropriate and current references were not always available, so while the board knew the processes, they were unable to provide adequate references for some teaching points.
4. Board mandates did not have any guidance regarding PRes and lacked specificity as per which of the training they should be engaging in.

Scheduling

1. QL 5A board scheduling was too short. 8 days was not enough and should be 3 weeks as a minimum.

QSP Process

1. Setting ground rules was a great way to start the board.
2. Scalaring the POs and EOs on the walls is very helpful.
3. If all members of the Board had DWAN access, using a SharePoint site would prevent having multiple versions of the document, facilitate research for referencing and would eliminate redundancies.
4. There needs to be a more defined process for achieving ‘approval’ for chapters 1 and 2 of the QSP prior to the Board moving on to Design.
5. Modularization of both the 5A and 5B will allow PRes and RegF members the flexibility in training. Future coursing should adhere to this format.

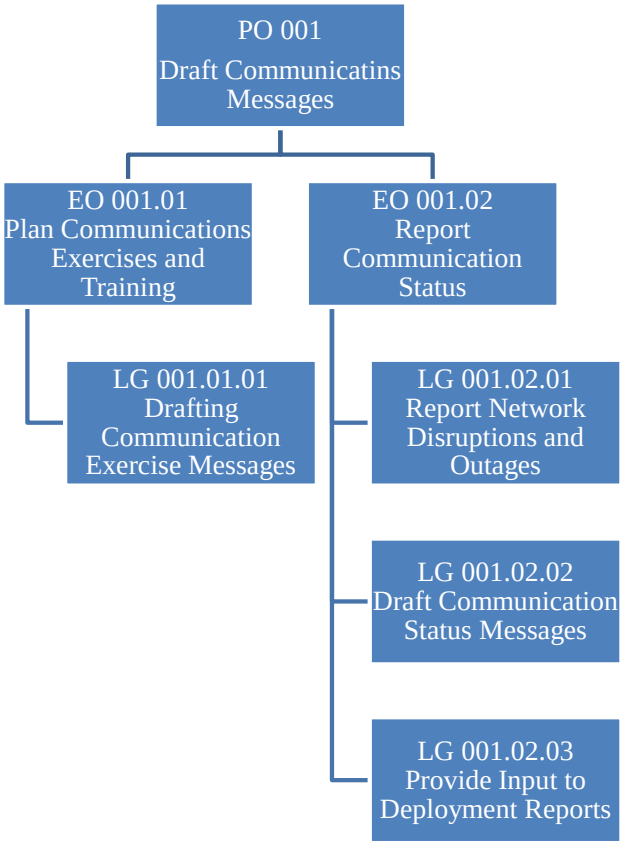
MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

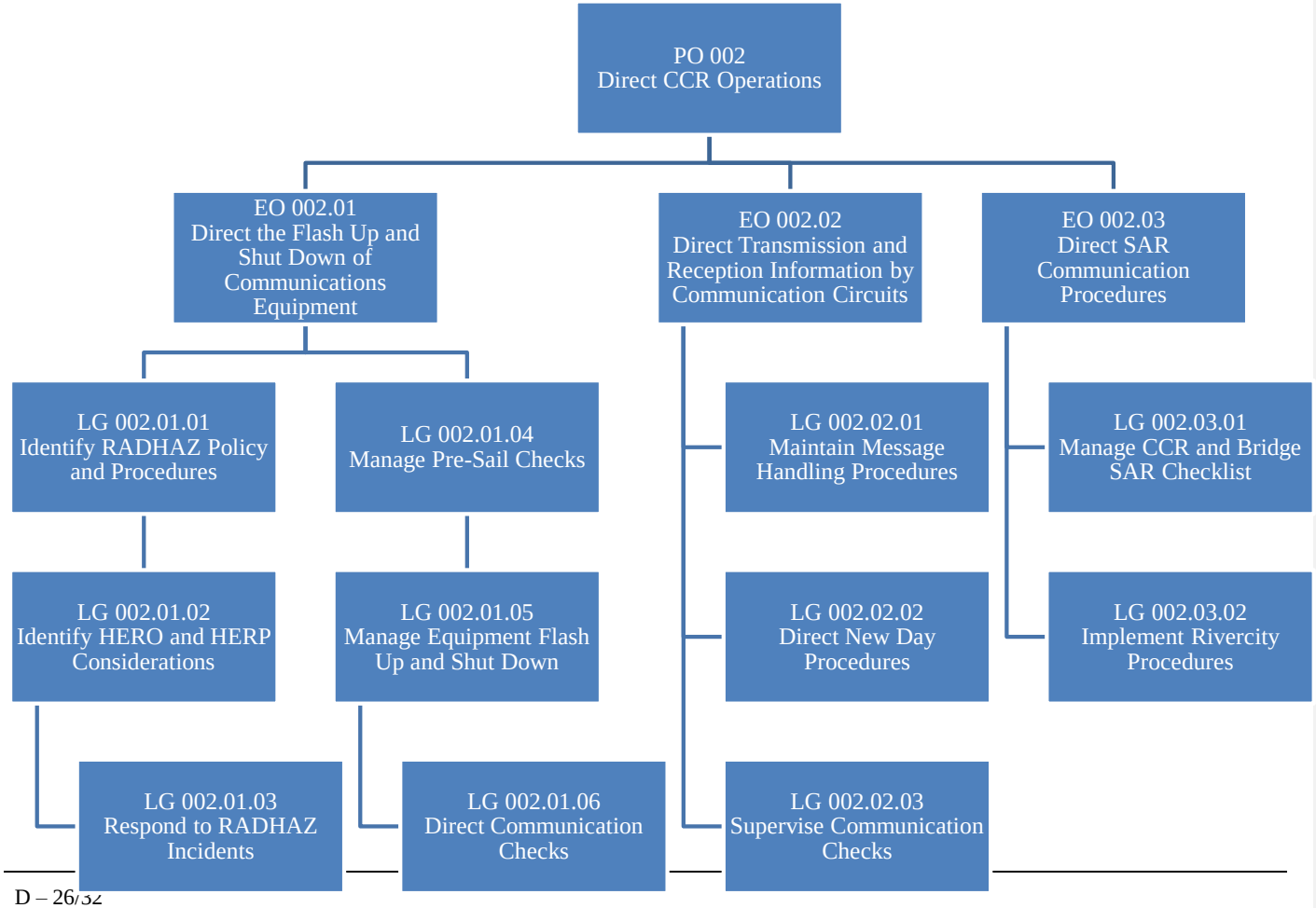
ANNEX D – TASK ANALYSIS DIAGRAM

This diagram represents the recreation or original task analysis (scalar) used to define the job.

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

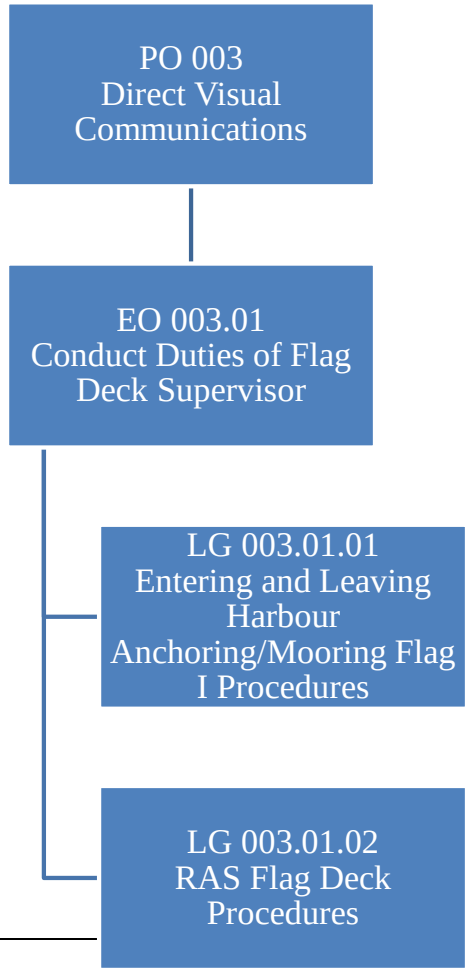


MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

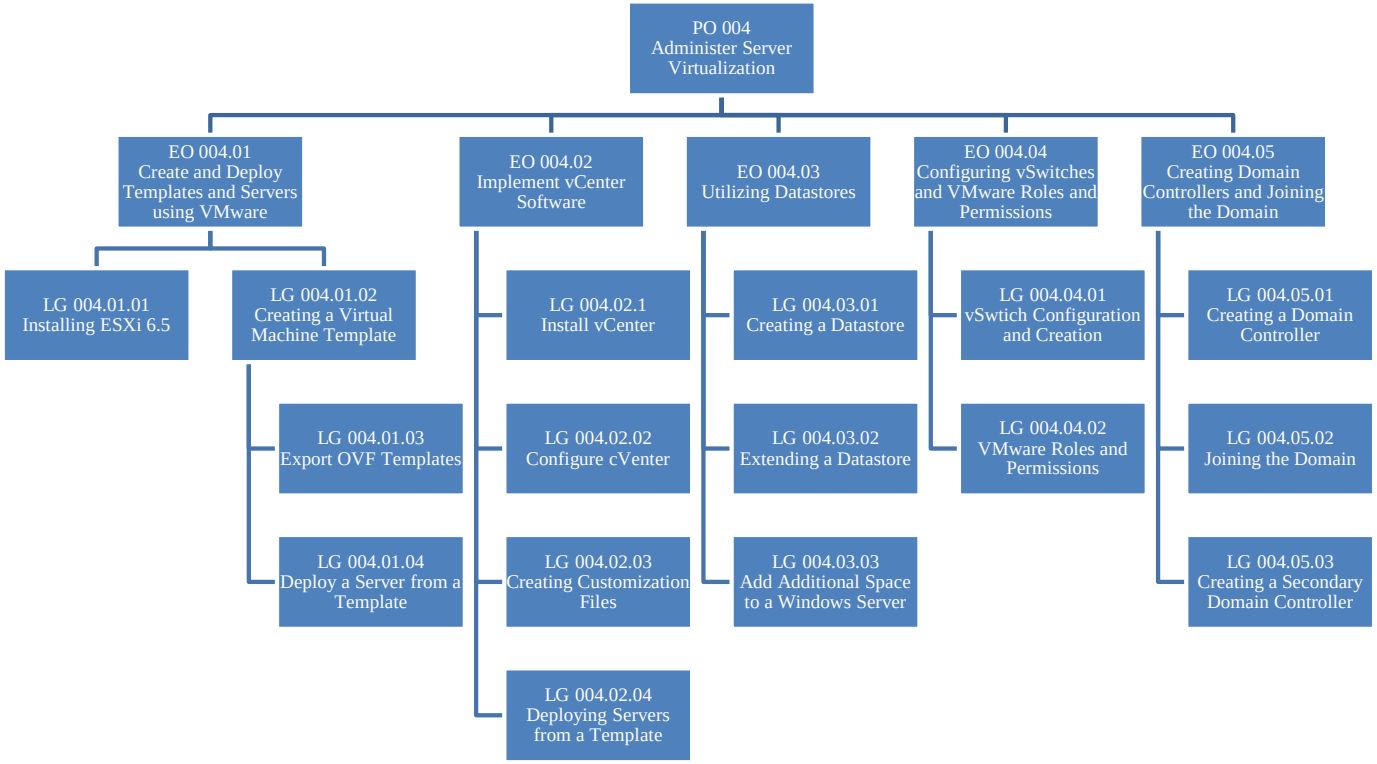


MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

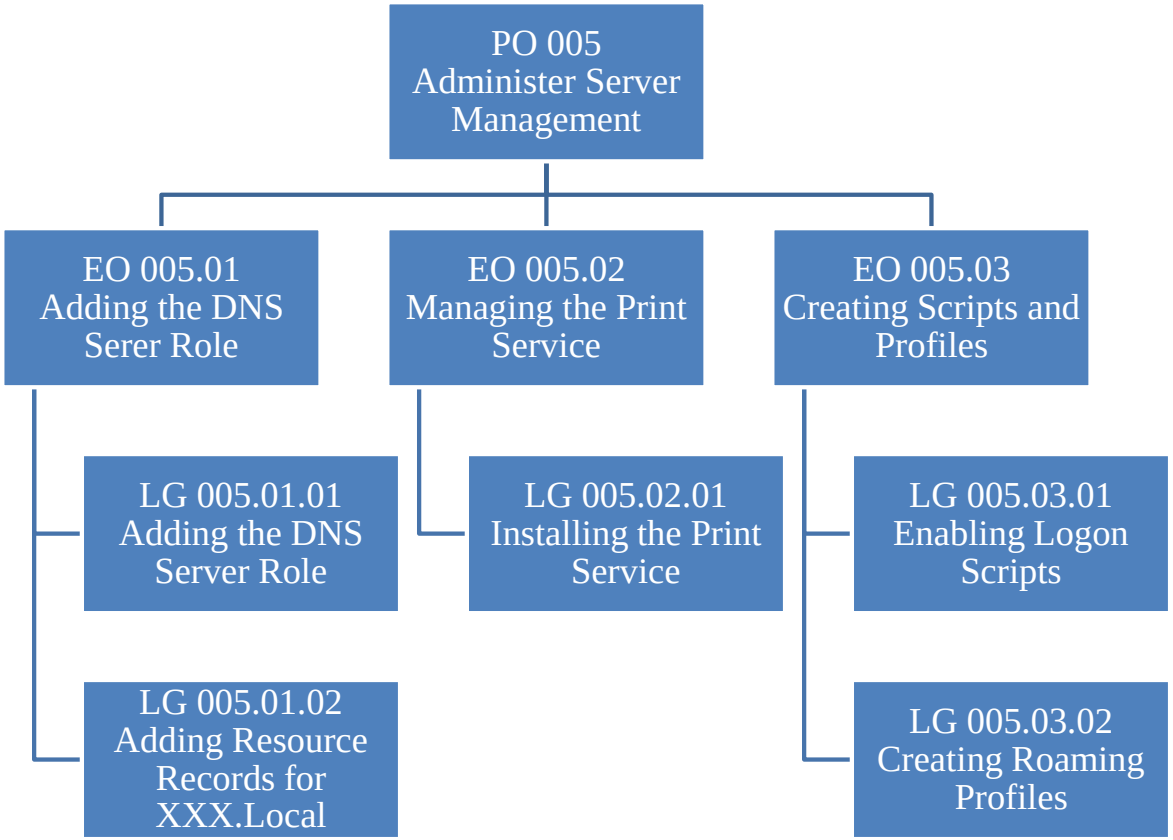
MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170



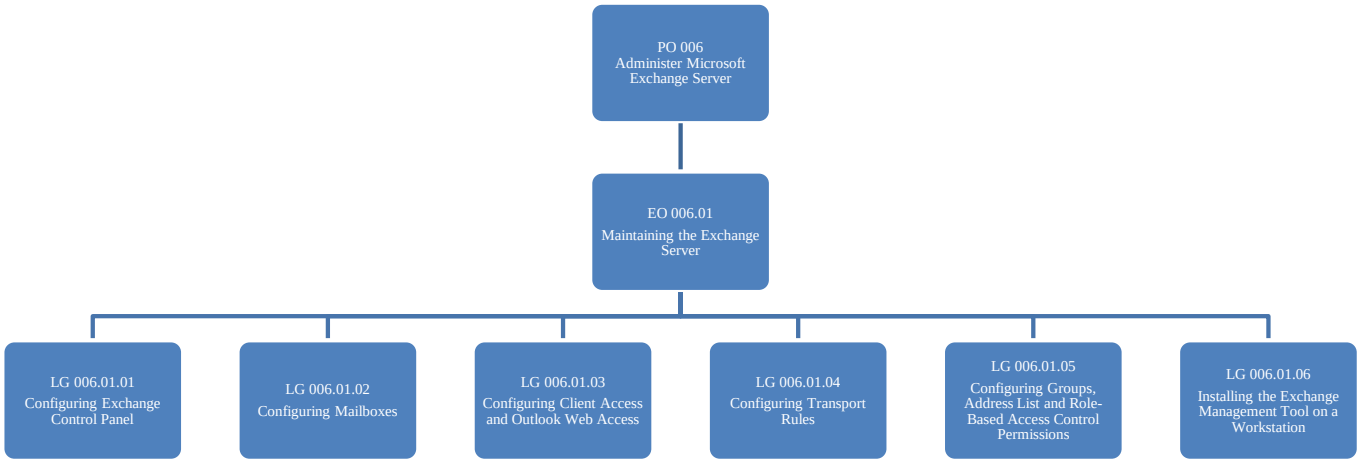
MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170



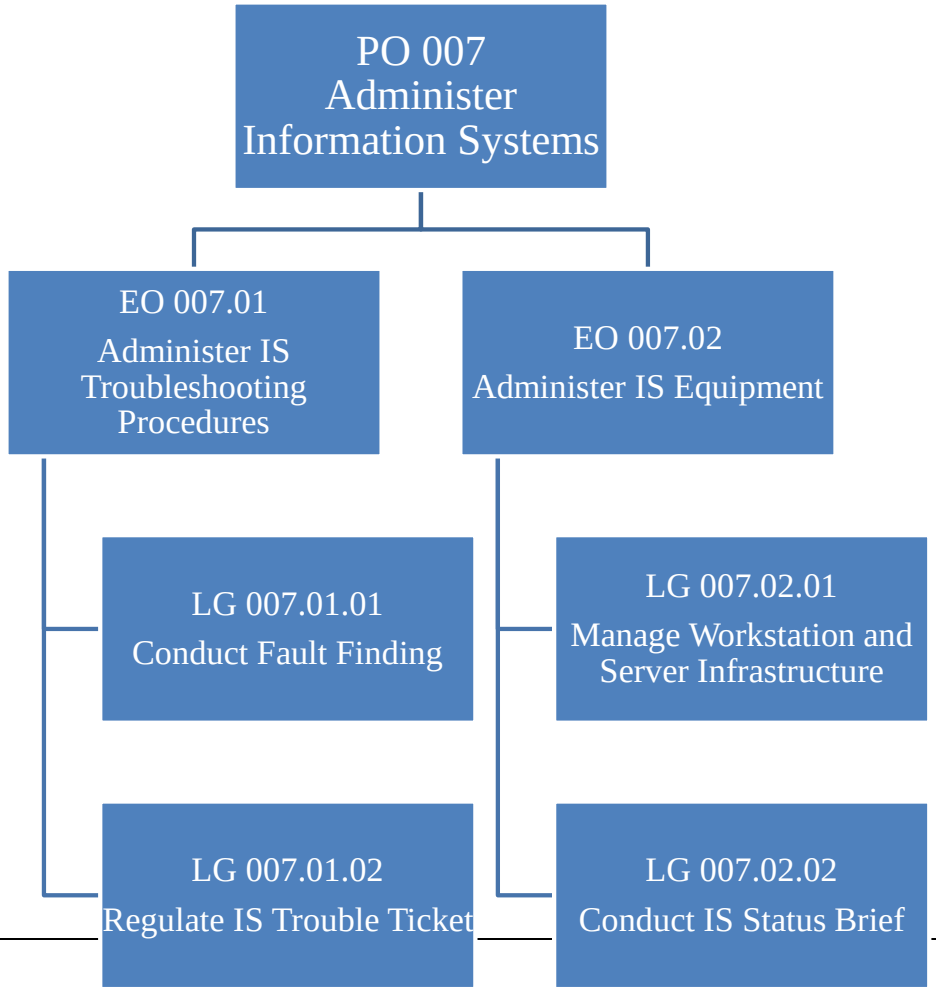
MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170



MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170



MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170



MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ANNEX E – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates. Personnel costs are IAW the Cost Factors Manual for the current fiscal year for all instructors identified in Chapter 1 - Required Course Staffing.

TD Costs	**# Trainees + Instructors /**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
Airfare (Return)	X	$0.00	n/a	$0.00	$0.00	$0.00
R&Q	X	$0.00	X	$0.00	$0.00	$0.00
Incidentals	X	$0.00	X	$0.00	$0.00	$0.00
Subtotal				**$0.00**	**$0.00**	**$0.00**
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	X	$0.00	X	$0.00	$0.00	$0.00
Classroom (square meters)	X	$0.00	X			
Computer requirements	X	$0.00	X	$0.00	$0.00	$0.00
IT Equip (Smart board, etc.)	X	$0.00	X	$0.00	$0.00	$0.00
DWAN connection (Setup)	X	$0.00	X	$0.00	$0.00	$0.00
Link Toolkit (on DVD)	X	$0.00	X	$0.00	$0.00	$0.00
Instructional Materials (Printed / Handouts etc.)	X	$0.00	X	$0.00	$0.00	$0.00
Subtotal				**$0.00**	**$0.00**	**$0.00**
Personnel Costs	**Quantity**	**Cost/day**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PO1	X	$0.00	X	$0.00	$0.00	$0.00
Civilian Instructor						
Subtotal				**$0.00**	**$0.00**	**$0.00**
Total Estimated Costs				**$0.00**	**$0.00**	**$0.00**

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ANNEX F – ASSESSMENT PLAN

This table summarizes assessment (EC & PC) details contained within this QSP.

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess trainee's competence in planning communications exercises and training.	Written / Practical	Trainee must obtain 13/13 SATs IAW EC 001.01 checklist.
EC 001.02	EO 001.02	To assess trainee's competence in reporting communication status.	Written / Practical	Trainee must obtain 27/27 SATs IAW EC 001.02 checklist.
PC 001	PO 001	To assess trainee's competence in drafting communications messages.	Written	Trainee must obtain 12/15 SATs IAW PC 001 checklist.
EC 002.01.01	EO 002.01	To assess trainee's competence in identifying RADHAZ policies and procedures.	Written	Trainee must obtain 8/10 SATs IAW PC 001 checklist.
EC 002.01.02	EO 002.01	To assess trainee's competence in managing pre-sail checks.	Written	Trainee must obtain 13/16 SATs IAW EC 002.01.02 checklist.
EC 002.02	EO 002.02	To assess trainee's competence in directing transmission and reception of information by communication circuits.	Written	Trainee must obtain 8/10 SATs IAW EC 002.02 checklist.
EC 002.03	EO 002.03	To assess trainee's competence in directing search and rescue communication procedures.	Written	Trainee must obtain 8/10 SATs IAW EC 002.03 checklist.
PC 002	PO 002	To assess trainee's competence in directing Communication and Control room operations.	Written	Trainee must obtain 16/20 SATs IAW PC 002 checklist.
EC 003.01.01	LG 003.01.01	To assess trainee's competence in conducting the duties of flag deck supervisor.	Written	Trainee must obtain 8/10 SATs IAW EC 003.01.01checklist.
EC 003.01.02	LG 003.01.02	To assess trainee's competence in conducting the duties of flag deck supervisor during a RAS.	Written	Trainee must obtain 8/10 SATs IAW EC 003.01.02checklist.
PC 003	PO 003	To assess trainee's competence in directing visual communications.	Written	Trainee must obtain 12/15 SATs IAW PC 003 checklist.
EC 004.01	EO 004.01	To assess trainee's competence in building ships servers to an operational state	Practical	Trainee must obtain 13/13 SATs IAW EC 004.01 checklist.

EC 004.04	EO 04.04	To assess trainee's competence in configuring vSwitches and VMware roles and permissions.	Practical	Trainee must obtain 14/14 SATs IAW EC 004.04 checklist.
EC 004.05	EO 004.05	To assess trainee's competence in creating domain controllers and joining the domain.	Practical	Trainee must obtain 14/14 SATs IAW EC 004.05 checklist.
PC 004	PO 004	To assess trainee's competence in administering server virtualization	Practical	Trainee must obtain 21/21 SATs IAW PC 004 checklist.
EC 005.01	EO 005.01	To assess trainee's competence in adding the DNS server role.	Practical	Trainee must obtain 6/6 SATs IAW EC 005.01 checklist.
EC 005.02	EO 005.02	To assess trainee's competence in managing the printer service.	Practical	Trainee must obtain 3/3 SATs IAW EC 005.02 checklist.
EC 005.03	EO 005.03.01	To assess trainee's competence in creating scripts and profiles	Practical	Trainee must obtain 5/5 SATs IAW EC 005.02 checklist.
PC 005	PO 005	To assess trainee's competence in administering server management	Practical	Trainee must obtain 7/7 SATs IAW PC 005 checklist.
EC 006.01	EO 006.01.01-EO 006.01.06	To assess trainee's competence in maintaining the exchange server	Practical	Trainee must obtain 5/5 SATs IAW EC 006.01 checklist.
PC 006	PO 006	To assess trainee's competence in administering Microsoft Exchange.	Practical	Trainee must obtain 5/5 SATs IAW PO 006 checklist
EC 007.01	EO 007.01	To assess trainee's competence in administering IS troubleshooting procedures.	Written / Practical	Trainee must obtain 14/14 SATs IAW EC 007.01 checklist.
EC 007.02	EO 007.02	To assess trainee's competence in administering IS equipment.	Written / Practical	Trainee must obtain 10/10 SATs IAW EC 007.02 checklist.
PC 007	PO 007	To assess trainee's competence in administering information systems.	Written / Practical	Trainee must obtain 13/13 SATs IAW PC 007 checklist.

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

In the event that a training methodology is determined as ineffective and/or inefficient, a Learning Support Request (LSR) may be submitted to correct the deficiency. Technology Enabled Learning (TEL) development opportunities are assessed IAW NPTORD 4500-84 following submission.

Title	Requirement Description	Lesson #	LSR #

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

ANNEX H – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
PO 001 – Draft Communications Messages											
LG 001.01.01	400	Classroom	CONF	1	13	0	13	-	-	1	Printer
EC 001.01	100	Classroom	CONF	1	13	0	13	-	-	1	Printer
LG 001.02.01	400	Classroom	CONF	1	13	0	13	-	-	1	Printer
LG 001.02.02	400	Classroom	CONF	1	13	0	13	-	-	1	Printer
LG 001.02.03	200	Classroom	CONF	1	13	0	13	-	-	1	Printer
LG 001.02.04	400	Classroom	CONF	1	13	0	13	-	-	1	Printer
EC 001.02	200	Classroom	CONF	1	13	0	13	-	-	1	Printer
PC 001	100	Classroom	CONF	1	13	0	13	-	-	1	Printer
PO 002 – Direct CCR Operations											
LG 002.01.01	500	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
EC 002.01.01	50	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
LG 002.01.02	400	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
EC 002.01.02	50	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
LG 002.02.01	400	Classroom	CONF	1	13	13	0	-	-	1	Printer
EC 002.02	50	Classroom	CONF	1	13	13	0	-	-	1	Printer
LG 002.03.01	400	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
EC 002.03	50	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
PC 002	50	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
PO 003 – Direct Visual Communications											
LG 003.01.01	200	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
EC 003.01.01	50	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
LG 003.01.02	400	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
EC 003.01.02	50	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
PC 003	100	Classroom	UNCLAS	1	13	13	0	-	-	1	Printer
PO 004 – Administer Server Virtualization											
LG 004.01.01	100	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer

MARPAC RDIMS# 1378818
CO NTDC (A) Approval – 14/12/2021
CNTG Approval – 08/08/2023
Version No: 3.0
Mite# 123169, 123170

LG 004.01.02	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 004.01	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 004.02.01	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 004.03.01	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 004.04.01	250	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 004.04	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 004.05.01	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 004.05	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PC 004	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PO 005 – Administer Server Management											
LG 005.01.01	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 005.01	100	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 005.02.01	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 005.02	100	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 005.03.01	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 005.03	100	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PC 005	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PO 006 – Administer Microsoft Exchange Server											
LG 006.01.01	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 006.01.02	250	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 006.01.03	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 006.01.04	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 006.01.05	250	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 006.01.06	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 006.01	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PC 006	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PO 007 – Administer Information Systems											
LG 007.01.01	500	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 007.01.02	500	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 007.01	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 007.02.01	850	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
LG 007.02.02	850	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
EC 007.02	150	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer
PC 007	200	NCITT	UNCLAS	1	13	1	0	-	-	1	Printer